

A Decomposition of Cryptocurrencies' Time Varying Systematic Risk Beta

Masud Alam, PhD,
Professor, Department of Economics,
Shahjalal University of Science and Technology, Bangladesh
E-mail: masudalam-eco@sust.edu

Extended Abstract

This study examines the dynamics of systematic risk in cryptocurrency markets by developing a dynamic asset pricing framework that moves beyond the static assumptions of traditional models. Unlike conventional approaches that rely on constant betas and exogenous macroeconomic factors, this project introduces a multifactor model with time-varying betas that specifically accounts for the unique characteristics of digital assets, including blockchain fundamentals, investor sentiment, and macro-financial conditions. Grounded in microeconomic theory and building upon the Arbitrage Pricing Theory, the study aims to identify cryptocurrency-specific risk factors—spanning technical, market, and behavioral dimensions—and decompose them into systematic and idiosyncratic components.

The analysis employs dynamic linear APT models estimated through state-space methods and the Kalman filter, augmented with machine learning techniques (XGBoost and LightGBM) and SHAP values to capture nonlinear relationships and provide interpretable factor importance rankings. To support this comprehensive analysis, the study constructs an extensive daily cryptocurrency dataset spanning from 2010 to 2025, covering 15 major digital assets—including Bitcoin, Ethereum, Solana, and XRP—that integrates diverse data sources such as blockchain metrics, market variables, macroeconomic indicators, and global risk measures.

Findings reveal that traditional static models, such as CAPM, exhibit low explanatory power and persistent negative alphas, underscoring their inadequacy in capturing cryptocurrency return dynamics. Time-varying beta analysis, however, uncovers rich patterns of risk evolution: Bitcoin shows gradual integration with traditional markets, Ethereum reflects DeFi and NFT-driven volatility, and Solana exhibits boom-bust sensitivity. Feature importance analysis indicates that network fundamentals—particularly transaction fees, transaction counts, and hashrate—dominate Bitcoin's systematic risk, explaining nearly 95% of predictive variation, while global financial stress indicators and monetary policy variables further modulate risk exposures.

We conclude that cryptocurrency risk is fundamentally endogenous, arising from the structural health of its underlying blockchain. However, it remains susceptible to exogenous shocks from traditional financial cycles, thereby positioning digital assets at the nexus of technological and financial systems. For investors, our findings contribute to the dynamic portfolio strategies that adapt to shifting cross-asset correlations and irregular surges in volatility. A static diversification approach is evidently inefficient. Significantly, this study shows that time-varying betas act as behavioral aspects to measure the risk-return mapping for the crypto ecosystem. The variation of betas effectively captures the complex interplay of technological innovation, speculative sentiment, and deepening market integration that characterizes this nascent asset class.

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I. Introduction

Traditional asset pricing theories propose that the anticipated excess return of a cryptocurrency or any asset is contingent upon its level of risk, commonly measured through factor's risk beta (Mankiw & Shapiro, 1984). Market risk beta, often referred to simply as "beta," measures an asset's sensitivity to movements in the overall market. Beta quantifies the systematic risk of an asset, which is the risk that cannot be diversified away by holding a diversified portfolio. According to the Sharpe (1964), Treynor (1962), and Lintner's (1965) version of the Capital Asset Pricing Model (CAPM), an expected risk premium of a cryptocurrency should be equal to the anticipated excess return of an aggregate market portfolio multiplied by its risk beta, where beta measured the covariance between the returns of cryptocurrency and those of the market (Wang & Chong, 2021). Also, in the single and multifactor index factor version of the CAPM, popularly known as Ross's (1976) Arbitrage Pricing Theory (APT), the expected risk premium of cryptocurrency comprises the summation of risk premiums linked to the macro-finance and equity risk factors, each multiplied by its corresponding beta, reflecting the cryptocurrency excess return's sensitivity to either market risk factor or other relevant risk factors included in the model (Ramos et al., 2021; Jo et al., 2020).

These theoretical frameworks enable investors to differentiate among various assets based on their level of risk, as quantified by their betas. Initially, in the formulation of CAPM and APT, investors make investment decisions assuming that,

- The market risk factor beta remains time-invariant,
- The relationship between the return of the individual cryptocurrency and the market risk factor or between the individual currency returns and the array of other risk factors is linear and
- The only source of risk factors is the market risk factor in the case of a single index factor model or a series of macro-finance or equity risk factors in the case of APT.

However, these theories did not specify which particular risk factors should be incorporated for the newly emerging crypto assets, potentially resulting in the omission of relevant sources

¹ Professor, Department of Economics, Shahjalal University of Science and Technology, Bangladesh
E-mail: masudalam-eco@sust.edu

of risk. Identifying these risk factors is particularly pertinent when analyzing the dynamics of cryptocurrency returns and constructing the optimum portfolio.

The theoretical studies regarding cryptocurrencies have identified several factors that potentially influence the fluctuation of cryptocurrency prices. One strand of literature is based on models about cryptocurrency's network effect (e.g., Biais et al., 2018; Pagnotta & Buraschi, 2018), emphasizing the price volatilities resulting from the positive spillover effects associated with network effects. Another body of literature concentrates on the production aspect of cryptocurrencies—specifically, the miners' predicament (Cong, He, and Li, 2021)—and observes that the dynamics of price volatility in cryptocurrencies is correlated with the rate of marginal cost structure of crypto mining. Additionally, a third set of studies focused on the linkage of cryptocurrency prices with the conventional asset classes, such as fiat currencies, stocks, bond, commodities (Liu et al., 2022), and crypto fundamental factors (Athey et al., 2016; Borri & Shakhnov, 2023). Moreover, there exists an expanding empirical literature on cryptocurrencies. Borri (2019) demonstrates that individual cryptocurrencies are susceptible to tail-risks in the crypto market. Makarov and Schoar (2020) ascertain periods of potential arbitrage opportunities across exchanges within cryptocurrency markets. However, it is noteworthy that, to the best of our knowledge, these bodies of literature are limited by their analytical frameworks, either through the assumption of constant risk beta over time or the presumption of linearity between the return of individual cryptocurrencies and the risk factors.

The inclusion of a single market risk factor and, therefore, the imposition of the time-invariant nature of estimated betas are unrealistic and are biased and overly restrictive estimations of betas in many studies. Starting from Fabozzi and Francis (1978) and Brooks, Faff, and Lee (1992), and recent studies by Fraser et al. (2004), Hrdlicka (2022), Umlandt (2023), and Raponi et al. (2020). Although the time-varying approaches to estimating time-varying risk betas have been applied extensively in conventional asset pricing literature with success in the literature of equity, bonds, commodities, real estate, and mutual funds, no study has as yet examined them about cryptocurrencies and provided the comparison to the constant market model beta. To fill this gap, this project aims to provide insights into how systematic risk varies over time. If beta is time-varying, it allows the model's flexibility to examine the statistical and economic significance of the non-linear relationship between cryptocurrency's risk premia, the market risk factor and risk beta, and the other risk factors.

This project attempts to propose, construct, and generalize a cryptocurrency-asset pricing model considering a multifactor APT and examine their empirical validity, in which we allow systematic risk beta to be time-variant. When we allow factors beta to be time-varying stochastic, we can overcome overly restrictive assumptions we observed in the traditional CAPM and APT because the proposed approach allows investors to consider the possibility of a non-linearity between crypto-specific excess returns and risk factors. Non-linearity is often absent in the current applications of APTs to explain crypto-asset returns. Moreover, the noble approach successfully tackles the effect of potential omitted variable bias, mainly caused by the model's failure to identify additional risk factors. In this context, our proposed approach considers crypto-specific technical, market, and investor sentiment factors in addition to macro-finance and traditional equity factors.

The motivation of the proposed arbitrage pricing model draws upon microeconomic theoretical foundations, particularly expected utility theory and market equilibrium, and the relationship between risk aversion, excess returns, and the risk factors in the Arbitrage Pricing Theory. The theory suggests that the expected risk premium of an asset can be modeled as a linear function of several systematic risk factors, also known as "macroeconomic factors" or "common factors." In the context of Arbitrage Pricing Theory, risk aversion influences investors' willingness to hold assets with different risk exposures. If investors become more risk-averse, they will demand a higher risk premium for bearing each unit of systematic risk, leading to an increase in expected returns across all assets. In crypto investing, investors consider the expected utility of various cryptocurrency choices, which accounts for both the potential returns and the associated risks. Risk-averse individuals weigh the potential for losses more heavily than potential gains, leading them to demand higher expected returns for riskier investments. Moreover, investors' attitudes towards risk shape their expectations of returns and play a crucial role in determining equilibrium in the cryptocurrency market.

Motivated by the microeconomic theory of uncertainty, this project attempts to integrate the conditional multifactor APT with a noble estimation approach to estimate the stochastic process for systematic risk beta. The proposed integration allows us to examine how the dynamics of the risk beta's conditional moments can influence the individual cryptocurrency's conditional expected return, variance, and covariance as functions of the conditional moments of risk beta. With the integration of time-varying risk betas and conditional moments, we expect to bring various crypto-specific uncertainty and risk factors independent of the macro-finance and equity factors. The objective is to decompose and examine whether the additional

risk factor is diversifiable or non-diversifiable. To achieve this objective, we attempt to construct a new conditional multifactor asset pricing model and provide economic and statistical justifications about including time-varying risk betas to explore if there is any extra systematic risk in explaining the dynamics of cryptocurrency prices.

Time-varying betas enable the analysis to accommodate non-linear dynamics in asset returns and risk and potential mis-specifications of risk sources. The primary contribution of this study lies in the decomposition of systematic risk within a multifactor APT framework with time-varying betas. By treating estimated beta as a random variable, this study introduces an extra source of risk components separate from the conventional risk factors. The variability and the variance decomposition of betas may capture other cryptocurrency-specific variables beyond conventional factors. For instance, we anticipate contributing to the literature by introducing new variables such as cryptocurrency active addresses, average transaction fee, block size, Google Trends, hash rate, mining profitability, currency-related tweets, and currency sent from addresses. These variables may summarize non-diversifiable stochastic risk components, diversifiable idiosyncratic factors, or a combination of diversifiable and non-diversifiable risk components. Finally, this study contributes to the empirical literature by examining the economic and empirical validity of the time-varying betas and newly proposed modified conditional multifactor APT.

II. Research Objectives

The objective of this project is

- Develop a cryptocurrency-specific asset pricing model that considers the limitations inherent in the CAPM and conditional APT. This model would allow for the analysis of non-diversifiable systematic risk with time-varying betas.
- Identify sources of uncertainty and crypto-specific variables independent of macro-finance and equity factors when risk betas are random.
- Decompose other sources of risk factors and crypto-specific variables into two components - time-varying diversifiable and non-diversifiable risk.
- Evaluate the theoretical and empirical validity of the proposed multi-factor APT if the constant risk beta hypothesis for the studied sample is rejected.

- Create a cryptocurrency data frame that includes a wide range of crypto-specific variables that were not previously considered.
- Determine which crypto-specific and other risk factors are non-diversifiable and require compensation for investors, as well as identify which time-varying risk factors are diversifiable.

III. Research Questions

- How can a cryptocurrency-specific multifactor asset pricing model with time-varying betas effectively capture the dynamic systematic risk exposures of cryptocurrencies?
- To what extent does the proposed multifactor asset pricing model outperform traditional static CAPM and APT models in explaining cryptocurrency excess returns?

In examining this model-specific research question, we focus on developing and validating a cryptocurrency-specific model with time-varying betas, addressing the project's objective to overcome the limitations of static CAPM and APT. The question emphasizes model performance in capturing dynamic risk exposures, which is central to our project's innovation. We also aim to identify and decomposition of crypto-specific risk factors, aligning with the project's goal to distinguish diversifiable and non-diversifiable risks. The identification and decomposition ensure a focus on factors unique to cryptocurrencies, independent of traditional macro-finance factors. So, we question what are the crypto-specific risk factors, independent of macro-finance and equity factors, that contribute to non-diversifiable systematic risk in cryptocurrency returns, and how can these factors be decomposed into time-varying diversifiable and non-diversifiable components?

In addition to our specific research questions, we set complementary research questions to explore specific crypto-related factors (e.g., network effects, mining costs, sentiment) and their time-varying nature, linking to portfolio construction, which is a key expected outcome. These complementary questions build on the novel risk factor (G_t) we theoretically introduced in the model, examining its contribution to explaining return volatility, which complements the main questions by testing the model's innovation. These questions also address the project's aim to assess non-linearity in the risk-return trade-off, ensuring the model captures complex dynamics and informs risk decomposition for portfolio optimization. Therefore, we examine

to what degree crypto-specific risk factors, such as network effects, mining cost dynamics, and investor sentiment, exhibit time-varying sensitivities in a multifactor APT framework, and how do these sensitivities influence portfolio construction for cryptocurrency investors? How does the incorporation of a novel risk factor derived from the stochastic disturbance term in the time-varying beta model enhance the explanatory power of cryptocurrency return volatility compared to models assuming constant risk betas? Finally, what is the empirical and economic significance of non-linear relationships between cryptocurrency returns and crypto-specific risk factors, and how do these relationships affect the identification of diversifiable versus non-diversifiable risk components in portfolio optimization?

IV. Significance of studying Time Varying Systematic Risk Beta

Examining cryptocurrency returns within the framework of the Capital Asset Pricing Model (CAPM) and Arbitrage Pricing Theory (APT) is significant due to cryptocurrencies' unique characteristics and the evolving nature of financial markets.

Firstly, cryptocurrencies represent a novel asset class with distinct risk-return profiles compared to traditional assets like stocks, bonds, and commodities. Understanding the drivers of cryptocurrency returns (Woebbeking, 2021) through models like CAPM and APT can provide insights into the underlying factors influencing their valuations and price dynamics. However, a systematic investigation of crypto-specific risk factors remains critical for investors navigating this emerging asset class.

Secondly, the cryptocurrency assets market is characterized by noticeable volatility and rapid price fluctuations, which make risk assessment and informed decision-making relatively difficult (Katsiampa, 2019). Applying time-varying beta frameworks within conventional models like the CAPM and APT offers a method to better evaluate risk-adjusted returns and to construct portfolios aligned with specific risk-return objectives.

Furthermore, blockchain technology and the market's decentralized nature, continuous operation, and comparatively limited regulatory oversight introduce distinct informational asymmetries. These asymmetries increase the utility of robust asset pricing models for analyzing returns and their linkages to fundamental risk drivers. Finally, as the cryptocurrency market matures and its integration with traditional finance deepens, the importance of applying these established theoretical frameworks grows increasingly relevant (Corbet et al., 2018).

By examining cryptocurrency returns within the framework of time varying betas, CAPM and APT, we aim to contribute to the ongoing development of theoretical models and practical tools for assessing risk and pricing assets in emerging and evolving markets.

Expected outcomes:

Aligned with our objectives, we attempt to propose and develop a variant of Capital Asset Pricing Model (CAPM), where we plan to integrate both CAPM and Arbitrage Pricing Theory and incorporate dynamic stochastic risk betas. Moreover, the study delivers the following outputs:

- The generalization of the dynamic linear CAPM and modification of the time-varying betas APT model to assist crypto investors in constructing optimal portfolios.
- Provides economic and financial rationale to justify the modified time-varying betas APT model and examining the portfolio implications of additional sources of risk factors within the cryptocurrency asset domain.
- It offers theoretical and empirical evidence against the restrictive assumption of constant betas in traditional asset pricing models.
- Development of an identification strategy to assess the statistical significance of non-linearity in explaining the risk-return trade-off and the presence of additional risk sources.
- Compilation of a comprehensive dataset encompassing various crypto-specific risk factors, spanning from crypto production, fundamental, and network factors to investor sentiment indices.
- Insights for crypto investors to distinguish whether the extra risk factors are time-varying diversifiable, idiosyncratic factors, or a combination of systematic and non-systematic components.
- Contribution to the ongoing debates surrounding modeling and the identification of risk factors in explaining the dynamics of cryptocurrency returns.
- Additionally, we aim to publish a peer-reviewed journal article in a Q1 journal documenting the time variation of systematic risk beta, shedding light on the potential impact of omitting crypto-specific risk factors in traditional CAPM or APT models, as evidenced in recent literature.

V. Literature Review

We focus on developing a cryptocurrency-specific asset pricing model with time-varying betas, identifying crypto-specific risk factors, decomposing risk into diversifiable and non-diversifiable components, and evaluating the model's theoretical and empirical validity. The research questions emphasize the effectiveness of the time-varying beta model, the identification of crypto-specific risk factors, and their implications for portfolio construction and risk decomposition. The econometric models include a multifactor dynamic APT with state-space representation and stochastic disturbance decomposition. Therefore, we identify five dimensions and segments of literature relevant to our project's research questions, aims, objectives, and econometric models. These five segments cover theoretical foundations of asset pricing theories and time-varying betas, crypto-specific risks, risk decomposition, econometric methods, and market dynamics.

Theoretical models of asset pricing such as the Capital Asset Pricing Model (CAPM) by Sharpe (1964) and the Arbitrage Pricing Theory (APT) by Ross (1976) established seminal paradigms for understanding market risk. These models conventionally assume static beta coefficients that quantify an asset's sensitivity to systematic risk. However, empirical inconsistencies and evolving market dynamics have prompted the development of models that account for time-varying risk exposures, notably the conditional CAPM by Jagannathan and Wang (1996) and the econometric advances introduced by Petris et al. (2009). Despite their robustness, these models have not been fully extended to the domain of cryptocurrencies, an asset class characterized by high volatility, 24/7 trading, and unique sources of systemic risk. It synthesizes the necessity of theoretical developments and identifies fundamental gaps that justify the development of a cryptocurrency-specific, dynamic asset pricing model.

Sharpe's (1964) Capital Asset Pricing Model (CAPM) establishes a linear relationship between an asset's expected return and its beta, a coefficient measuring sensitivity to overall market risk. This model, grounded in the assumptions of mean-variance optimization and frictionless markets, offers an elegant theoretical foundation. Its practical applications, however, is constrained by the assumption of a static beta, a significant limitation in volatile trading environments. Furthermore, the single-factor specification of CAPM fails to capture additional macroeconomic or idiosyncratic risks, which proven a critical shortfall for emerging digital asset classes like cryptocurrencies.

In methodological contributions, Ross's (1976) Arbitrage Pricing Theory addresses the first limitation by introducing a multifactor framework. This APT model accommodates multiple

systematic risk sources, each with an independent beta coefficient, and offers flexibility by not pre-specifying the risk factors. However, APT framework largely retains the conventional assumptions of constant betas and linear return-generating processes. These static assumptions frequently established oversimplify the complex and rapidly shifting risk exposures inherent in highly volatile markets. For cryptocurrencies, where valuations are highly sensitive to regulatory news, network factors, technological innovation, and investor sentiment—this rigidity can substantially diminish the models' explanatory power.

The shortcomings of static beta models led to the development of time-varying frameworks. Jagannathan and Wang (1996) advanced this literature by introducing a conditional CAPM, wherein both betas and risk premia vary over time based on macroeconomic indicators. Their empirical work demonstrated that allowing betas to change with business cycle conditions improved the explanatory power of asset returns, especially in equity markets. This dynamic approach aligns conceptually with the objective of modeling cryptocurrency risk exposures, although their implementation remains rooted in traditional economic instruments that may not translate effectively to digital asset markets. Their reliance on business cycle proxies underscores a limitation when addressing the more immediate and technologically driven fluctuations typical in crypto markets.

In a recent methodological support for modeling dynamic beats, Petris et al. (2009) provides a comprehensive treatment of dynamic linear models (DLMs) within a Bayesian state-space framework. These models facilitate time-varying parameter estimation, including betas, through stochastic processes and Kalman filtering techniques. While Petris et al. did not apply their methods to financial markets directly, their framework is particularly well-suited to modeling non-stationary and heteroskedastic return series, such as those observed in cryptocurrency markets. The state-space approach offers both flexibility and precision, making it a strong candidate for extending traditional asset pricing models into more complex and volatile environments.

While we acknowledge these contributions, we also identify several critical gaps in the literature, especially in the context of cryptocurrency markets. First, foundational models like CAPM and APT were developed for traditional asset classes and do not account for the idiosyncrasies of digital assets, such as decentralization, network-specific shocks, or protocol-level risks. The absence of crypto-specific beta estimates and risk factors limits the direct applicability of these models. Second, while the conditional CAPM addresses time variation in

risk exposure, it assumes macroeconomic predictors that do not map neatly onto the rapidly shifting conditions in the crypto ecosystem. Crypto specific market sentiment such as tweets and google trends, blockchain activity, and network congestion, for example, are more relevant to cryptocurrencies than GDP growth or interest rate cycles. Third, while APT’s multifactor structure is theoretically flexible, there is little empirical guidance on identifying or validating crypto-specific systematic factors, such as mining difficulty or developer activity. From the perspective of methodological concerns, the decomposition of stochastic disturbances in time-varying beta models is also not addressed adequately in studying the dynamics of cryptocurrency asset returns. Seminal works—including those of Jagannathan and Wang (1996) or Petris et al. (2009)—do not address the derivation of new, endogenous risk factors from the residual components of state-space models.

In this paper, we aim to contribute by incorporating novel risk components derived from model errors, potentially revealing latent sources of systemic risk unique to cryptocurrencies. The existing literature also lacks practical applications of state-space modeling specifically tailored to crypto assets, despite their suitability for capturing non-linearities, jumps, and regime shifts in returns. We propose a novel framework for modeling cryptocurrency returns using a multifactor, time-varying beta model estimated within a Bayesian state-space architecture. The first research question—how a cryptocurrency-specific model with time-varying betas compares to static models in explaining excess returns—directly engages with the assumptions of Sharpe (1964) and Ross (1976) and extends the logic of Jagannathan and Wang (1996) to a new asset class. By integrating Petris et al.’s (2009) techniques, the model enables precise estimation of dynamic beta paths in response to rapidly evolving market conditions. The second research question—whether a risk factor derived from stochastic disturbance decomposition enhances explanatory power—addresses a novel aspect not explored in previous models. This innovation extends Ross’s APT by offering a new mechanism for identifying systemic risks, derived not from observable macroeconomic variables but from the structure of the model itself. It builds on the flexibility of the APT while grounding it in a robust state-space econometric framework.

From a theoretical perspective, this research integrates and extends the seminal literature by adapting traditional models to the unique structure of the cryptocurrency market. Sharpe’s (1964) CAPM provides the foundational logic for systematic risk pricing, which this research modifies by allowing beta to evolve over time. Ross’s (1976) APT informs the multifactor design, permitting the inclusion of crypto-specific drivers of return. Jagannathan and Wang

(1996) justify the theoretical necessity of dynamic betas, while Petris et al. (2009) provide the tools to estimate such models in practice. Together, these elements combine into a coherent framework tailored to the cryptocurrency context, advancing asset pricing theory beyond its traditional boundaries. While Sharpe and Ross address broad asset classes in static settings, this study concentrates on the dynamic behavior of cryptocurrencies, a market largely ignored in prior models. Similarly, while Jagannathan and Wang addressed time variation in equities, the present research applies this to a digital asset class where betas may shift daily—or even hourly. Finally, while Petris et al. offer generic econometric tools, their application to crypto markets introduces methodological rigor to a previously underexplored domain.

VI. Methodology:

Data

In the empirical analysis, we begin by examining three dimensions of cryptocurrency data alongside crypto-specific risk factors. These dimensions include cryptocurrency-specific network and production factors, fundamental risk factors pertaining to cryptocurrencies, and cryptocurrency market investor sentiment factors. The network and production-related variables encompass metrics such as the daily number of blockchain transactions, average block size, and the count of unique sending addresses. To further characterize network activity, we also include the daily number of unique addresses interacting either as sender or receiver, the proportion of total coins held by the top 100 richest addresses, average mining difficulty, and mining profitability. Additional measures include the total value of coins transferred in U.S. dollars, average block time in minutes, and the average hashrate, which represents computational power deployed per day.

We also examine a set of fundamental risk factors, which reflect both price momentum and structural volatility. These include 7-day, 30-day, and 90-day momentum indicators, as well as the TRIX (Triple Exponential Average) of average prices, all measured daily in USD. To capture market fluctuations, we assess the volatility in market capitalization and the average transaction value. The average transaction fee per day is also considered as a proxy for network congestion and user demand. The investor sentiment is represented using two widely recognized proxies: the number of daily Bitcoin-related tweets and the frequency of searches for “Bitcoin” as reported by Google Trends. Subsequently, we integrate the cryptocurrency dataframe with conventional macro-finance data and a diverse range of equity risk factors.

These variables serve as indirect indicators of public interest and speculative attention, which are known to influence short-term price behavior in cryptocurrency markets.

Our dataset spans from July 2010 to April 2025, providing daily observations for 15 actively traded cryptocurrencies: Bitcoin (BTC), Bitcoin Cash (BTCcash), Bitcoin SV (BTCSV), Cardano (CARDANO), Dash (DASH, also known as Digital Cash), Dogecoin (DOGE), Ethereum (ETH), Ethereum Classic (ETHCL), Litecoin (LITECOIN), Monero (MONERO), Solana (SOLANA), TRON (TRON), Vertcoin (VERT), Ripple's XRP, and Zcash (ZCASH). The selection of these cryptocurrencies is based on the availability of both network fundamentals and market risk factors. However, while Bitcoin data offers the longest continuous record, beginning in 2010, the coverage for other digital assets varies depending on the year they were listed and began trading on exchanges. This results in an unbalanced panel, with differing starting points for each cryptocurrency depending on their market entry.

Bitcoin (BTC), the first and most widely recognized cryptocurrency, serves as a decentralized digital currency and store of value, operating on a proof-of-work blockchain. Bitcoin Cash (BCH) originated from a 2017 hard fork of Bitcoin, primarily to enhance scalability through an increased block size, thereby facilitating faster and lower-cost transactions. A subsequent fork of Bitcoin Cash, Bitcoin SV (BSV), further pursued scalability and expanded data capacity with the explicit objective of restoring Bitcoin's original protocol for enterprise applications. In contrast, Cardano (ADA) employs a research-oriented, proof-of-stake framework to construct a blockchain platform supporting smart contracts and decentralized applications, with a foundational emphasis on sustainability and scalability.

Dash (DASH), an acronym for "Digital Cash," prioritizes rapid and inexpensive transactions complemented by optional privacy features, operating on a hybrid consensus mechanism. Initially conceived as a meme, Dogecoin (DOGE) has secured a notable position in the market due to its minimal transaction fees and strong community support, despite its simple proof-of-work architecture. Ethereum (ETH) remains the predominant decentralized platform for smart contracts and dApps, powered by its native currency, Ether, and is undergoing a significant transition from proof-of-work to a proof-of-stake consensus model.

Ethereum Classic (ETC) persists as the original, unaltered Ethereum blockchain following a historical fork, upholding a principle of absolute immutability. Often characterized as 'silver to Bitcoin's gold,' Litecoin (LTC) is a cryptocurrency engineered for swifter block generation and

reduced transaction fees. Monero (XMR) distinguishes itself through a strong privacy orientation, utilizing advanced cryptography to ensure fully opaque transactions and user anonymity.

Solana (SOL) utilizes a unique hybrid proof-of-history consensus to achieve high throughput, supporting decentralized applications with notably low transaction costs. The TRON (TRX) network is designed to decentralize digital content entertainment, enabling direct creator-to-audience engagement via smart contracts. Vertcoin (VTC) is a cryptocurrency explicitly designed to resist specialized ASIC mining, aiming to preserve mining accessibility and network decentralization. XRP (XRP), developed by Ripple Labs, is designed for fast, low-cost cross-border payments, functioning as a bridge currency for financial institutions. Zcash (ZCASH) is a privacy-centric cryptocurrency that uses zero-knowledge proofs to offer optional shielded transactions, balancing transparency and anonymity. The inclusion of the daily data on these cryptocurrencies collectively represents a spectrum of use cases, from payments and privacy to smart contracts and decentralized ecosystems, each contributing to the evolving landscape of digital finance.

The dataset utilized in this study is sourced from multiple platforms: Firstly, proprietary data is obtained from Bloomberg terminal, a global provider of economic and financial market data and the Center for Research in Security Prices, LLC (CRSP) which maintains the most comprehensive collection of security price, return, and volume data for the NYSE, AMEX and NASDAQ stock markets. Additionally, crypto-specific volume, trading, and liquidity data are sourced from <https://coinmarketcap.com>, while cryptocurrency network, production, and sentiment data comes from <https://bitinfocharts.com> through a premium paid API plan and web scraping methodology. Secondly, public sources such as investing.com, FRED St. Louis, and historical data for the Cboe VIX® Index and other volatility indices, Fama-French equity factors are utilized to frame a compact dataset.

Econometric Model: Time Varying Beta Model

For $i = 1, \dots, n$ cryptocurrencies and the period of t , we consider the vector of cryptocurrency returns $r_t = (r_{t1}, \dots, r_{tn})'$, the market return r_{mt} and the return on a risk free asset rf_t . Thus the vector of excess returns on n cryptocurrencies is defined as $ER_{i,t} = [r_{it} - rf_t]$ and the excess

market returns is $F_{m,t} = r_{mt} - rf_t$, and let the vector of other risk factors $F_1 \dots F_k$. Therefore, a multifactor version of dynamic APT model (DL-APT) can be written as:

$$\begin{aligned}
(1) \quad & ER_{i,t} = \alpha_{i,t} + \beta_{m,i,t} F_{m,t} + \sum_{j=1 \dots k; i \neq j} \gamma_{i,t} F_{j,t} + \varepsilon_{i,t} \text{ with } \varepsilon_i \sim N(0, \sigma) \\
(2) \quad & \alpha_{i,t} = \alpha_{i,t-1} + w_{i,t,1} \text{ with } w_{i,t,1} \sim N(0, \vartheta_1) \\
(3) \quad & \beta_{i,t} = \beta_{i,t-1} + w_{i,t,2} \text{ with } w_{i,t,2} \sim N(0, \vartheta_2) \\
(4) \quad & \gamma_{i,t} = \gamma_{i,t-1} + w_{i,t,3} \text{ with } w_{i,t,3} \sim N(0, \vartheta_3)
\end{aligned}$$

where α, β , and γ are $n - dimensional$ vectors. Unlike static APT market beta, our model confirms the coefficient of market risk factor $\beta_{m,i}$'s and the coefficients of other risk factors $\gamma_{i,t}$ are time varying. At first stage, our primary interest is on the $\beta_{m,i}$, which measures the response of asset i 's excess returns to the changes of the aggregate market returns, which is measured by the MSCI World Index and crypto 30 index. Following Petris et al. (2009) and Holmes, Scheuerell, and Ward, (2020), we define $\theta_t = [\alpha_t, \beta_t, \gamma_t]^T$, $H_t = I$, $w_t =$

$[w_{t,1}, w_{t,2}, w_{t,3}]^T$ and $F_t = [1, F_{m,t}, F_{j,t}]$ for all t , and $\Lambda = \begin{bmatrix} \vartheta_1 & 0 & 0 \\ 0 & \vartheta_2 & 0 \\ 0 & 0 & \vartheta_3 \end{bmatrix}$. So, we can write

the equations (1)-(4) as a state-space representation:

$$\begin{aligned}
ER_t &= (F_t^T \otimes I_n) \theta_t + \varepsilon_t \text{ with } \varepsilon_t \sim N((0, \sigma); \\
\theta_t &= (H_t \otimes I_n) \theta_{t-1} + w_t \text{ with } w_t \sim MVN(0, \Lambda);
\end{aligned}$$

Time Varying Beta: Theoretical Foundation

We assume that the stochastic process $\{\gamma_{i,t}\}$ which is a measure of the sensitivity of cryptocurrency i 's excess returns to a factor F is stationary and applying the Wold's decomposition, $\{\gamma_{i,t}\}$ can be written as,

$$\gamma_{i,t} = \gamma_i + \sum_{j=0}^{\infty} \varphi_j \delta_{i,t-j}$$

where $\varphi_0 = 1$ and $\delta_{i,t}$ is a white noise process, uncorrelated with the idiosyncratic noise, and orthogonal to the factor F . To derive the case where time varying risk beta can represent

additional sources of risk, we consider two currencies i and j such that $\gamma_i = \gamma_j$ and we let γ_i to be time varying random coefficient model and γ_j is the constant. If the assumption of same idiosyncratic risk factor holds for both currencies, we have the following:

$$\mu_{i,t} = \mu_{j,t}, \text{var}_{t-1}(R_{it}) > \text{var}_{t-1}(R_{jt}), \text{cov}_{t-1}(R_{it}, F_t) = \text{cov}_{t-1}(R_{jt}, F_t)$$

When two currencies exhibit the same expected return but currency i demonstrates higher return volatility in comparison to currency j , investors, possessing all available information, are likely to differentiate the currency with higher volatility. Given that investors receive equivalent expected returns, they will adjust their portfolios by reallocating their investments from currency i to currency j . The increased demand for currency j will result in an upward movement of its price, while the reduced demand for currency i will cause its price to decline. Consequently, the expected return from currency i is expected to exceed that of currency j . This portfolio adjustment process continues until the market achieves equilibrium. Therefore, if a currency with higher variance is perceived as riskier, a risk premium must exist, as measured by the time-varying beta. According to the Arbitrage Pricing Theory (APT) and the Capital Asset Pricing Model (CAPM), investors demand a higher expected return to hold riskier currency and $\mu_{i,t} \neq \mu_{j,t}$ as long as their variances differ.

With the arguments we presented above, a modified APT can be develop following Hildreth and Houck (1968) stochastic process model as

$$\gamma_{i,t} = \gamma_i + \theta_{it}$$

where θ_{it} is the white noise, and next, we decompose the stochastic disturbance term $\theta_{i,t}$. Let a decomposition of $\theta_{i,t} = \sigma_{\theta i} \tau_t$, where τ_t is the $\tau_t \sim i.i.d. (0,1)$. Therefore, excess return of a cryptocurrency i is,

$$ER_{i,t} = \alpha_{i,t} + \beta_{m,i,t} F_{m,t} + \gamma_i F_t + \sigma_{\theta i} \tau_t F_t + \varepsilon_{i,t}$$

$$ER_{i,t} = \alpha_{i,t} + \beta_{m,i,t} F_{m,t} + \gamma_i F_t + \sigma_{\theta i} G_t + \varepsilon_{i,t}$$

where $\tau_t F_t = G_t$ and the two factors are F_t and G_t . So, the conditional distributions of the cryptocurrency returns are:

$$\alpha_{it} = \beta_{m,i} \lambda_{mt} + \gamma_i \Omega_t + \sigma_{\theta i} \lambda_{1t} + \lambda_0$$

Where λ_{1t} is the risk premium for the newly decomposed factor G_t representing the extra source of risk premium in addition to market and all other relevant macro-finance and equity

factors. We attempt to prove the empirical and economic validity of the newly constructed factor in our subsequent empirical estimation.

VII. Results

Summary Statistics for Cryptocurrency Returns

The primary dataset comprises daily returns for 15 actively traded cryptocurrencies, spanning from July 2010 to April 2025, resulting in an unbalanced panel due to varying market entry dates for each asset. Bitcoin (BTC) provides the longest continuous record, starting in July 2010, while other cryptocurrencies, such as Bitcoin Cash (BTCcash), Bitcoin SV (BTCSV), Cardano, Dash, Dogecoin (DOGE), Ethereum (ETH), Ethereum Classic (ETHCL), Litecoin, Monero, Solana, TRON, Vertcoin (VERT), XRP, and Zcash, have shorter time series depending on their initial listing on exchanges. Table 1 presents the summary statistics of daily returns for these cryptocurrencies, including mean, standard deviation, median, minimum, maximum, skewness, kurtosis, and the number of observations (N). These statistics provide insights into the distributional properties of cryptocurrency returns, which are critical for modeling time-varying risk exposures in asset pricing frameworks.

Table 1. Summary Statistics of Daily Cryptocurrency Returns

Cryptocurrency	Mean	Std. Dev.	Median	Min	Max	Skewness	Kurtosis	N
BTC	0.0027	0.0381	0.0016	-0.4397	0.3830	-0.0626	14.5261	5,412
BTCcash	0.0007	0.0577	-0.0004	-0.3792	1.0913	4.5852	81.0801	2,845
BTCSV	-0.0004	0.0467	-0.0009	-0.2912	0.4263	0.7166	12.3842	2,358
CARDANO	0.0013	0.0513	0.0000	-0.2406	0.5660	1.4212	14.6556	2,779
DASH	0.0007	0.0662	0.0002	-2.1717	1.2570	-5.6054	331.4614	4,096
DOGE	0.0015	0.0621	0.0000	-0.6366	1.2324	3.7938	58.9914	4,162
ETH	0.0021	0.0576	0.0011	-0.8991	0.8433	-1.0470	44.8192	3,794
ETHCL	0.0010	0.0514	-0.0004	-0.2661	0.8303	2.1107	29.2966	3,213
LITECOIN	0.0017	0.0508	0.0003	-0.5125	0.6278	1.5772	23.1898	4,683
MONERO	0.0013	0.0463	0.0016	-0.2614	0.5113	0.8218	10.9921	3,994
SOLANA	0.0028	0.0534	0.0000	-0.4126	0.2848	0.0205	5.4929	1,857
TRON	0.0018	0.0511	0.0000	-0.3483	0.6408	2.4505	30.2428	2,738
VERT	-0.0009	0.0754	-0.0043	-0.4397	0.9009	2.3788	21.5631	3,460
XRP	0.0011	0.0648	0.0000	-0.5682	0.9317	1.7103	25.8175	4,260

ZCASH	-0.0025	0.0800	-0.0004	-2.8900	0.7357	-18.1979	616.8995	3,117
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Note: This table presents summary statistics for daily returns of 15 cryptocurrencies from July 2010 to April 2025, forming an unbalanced panel due to varying market entry dates. Bitcoin has the longest record (5,412 observations), while Solana has the shortest (1,857). Mean values of returns range from -0.0025 (Zcash) to 0.0028 (Solana), with high standard deviations varies from 0.0381 to 0.0800 reflecting significant volatility. Median returns are often near zero, indicating clustering of small returns. Skewness and high kurtosis (5.4929–616.8995) highlight non-normal distributions with fat tails, underscoring the need for dynamic asset pricing models to capture time-varying risk exposures in cryptocurrency markets.

The mean daily returns range from -0.0025 (Zcash) to 0.0028 (Solana), reflecting the heterogeneity in average performance across cryptocurrencies. Most assets exhibit positive mean returns, with Solana, Bitcoin, and Ethereum showing the highest at 0.0028, 0.0027, and 0.0021, respectively, while Bitcoin SV, Vertcoin, and Zcash display negative means, suggesting periods of underperformance. Standard deviations are notably high, ranging from 0.0381 (Bitcoin) to 0.0800 (Zcash), underscoring the elevated volatility characteristic of cryptocurrency markets. This volatility is further evidenced by the wide range between minimum and maximum returns, with extreme losses (e.g., -2.8900 for Zcash) and gains (e.g., 1.2570 for Dash), highlighting the potential for significant price swings.

The median returns, often close to zero (e.g., 0.0000 for Cardano, Dogecoin, Solana, TRON, and XRP), indicate that daily returns frequently cluster around small positive or negative values, consistent with the high-frequency nature of the data. Skewness varies considerably, with Zcash exhibiting extreme negative skewness (-18.1979), suggesting a pronounced left tail, while Bitcoin Cash (4.5852) and Dogecoin (3.7938) show strong positive skewness, indicating right-tailed distributions. Kurtosis values are uniformly high, ranging from 5.4929 (Solana) to 616.8995 (Zcash), with most cryptocurrencies exceeding the normal distribution's kurtosis of 3, confirming the presence of fat tails and extreme return events. Bitcoin, with 5,412 observations, has the longest time series, while Solana, with 1,857 observations, reflects its later market entry.

These distributional characteristics—high volatility, non-normal skewness, and leptokurtic returns—justify the need for dynamic asset pricing models that account for time-varying risk exposures, as static models like CAPM or APT may fail to capture the complex dynamics of cryptocurrency returns. The unbalanced panel, with varying observation counts (e.g., 1,857 for Solana to 5,412 for Bitcoin), necessitates robust econometric techniques, such as state-space models, to handle missing data and model time-varying parameters effectively. The summary statistics provide a foundation for empirical analyses aimed at identifying crypto-specific risk

factors and evaluating their impact on portfolio construction within the proposed time-varying beta framework.

The summary table of daily cryptocurrency returns reveal significant economic and financial implications that underscore the unique risk-return dynamics of this emerging asset class. The high volatility, evidenced by standard deviations ranging from 0.0381 (Bitcoin) to 0.0800 (Zcash), reflects the decentralized and speculative nature of cryptocurrency markets, which are influenced by factors such as regulatory uncertainty, technological advancements, and investor sentiment. Mean returns vary from a negative value of -0.0025 for Zcash to a positive 0.0028 for Solana. These variations suggest heterogeneous economic performance across cryptocurrencies and potentially driven by differing use cases, adoption rates, and market maturities. The pronounced skewness (e.g., -18.1979 for Zcash, 4.5852 for Bitcoin Cash) and elevated kurtosis (up to 616.8995 for Zcash) indicate frequent extreme price movements, highlighting the presence of tail risks that challenge traditional asset pricing models like CAPM and APT. These characteristics necessitate econometric models that account for time-varying risk exposures, as static beta assumptions may fail to capture the dynamic economic forces—such as network effects or mining cost fluctuations—driving cryptocurrency returns. Subsequent empirical analyses should prioritize identifying crypto-specific risk factors and testing their systematic versus idiosyncratic nature, using state-space models to estimate time-varying betas that reflect these economic realities.

From an investment perspective, the summary statistics inform portfolio construction and risk management strategies in cryptocurrency markets. The wide range of returns (e.g., -2.8900 to 0.7357 for Zcash) and high volatility suggest significant opportunities for returns but also substantial risks, requiring investors to differentiate between diversifiable and non-diversifiable risk components. The near-zero median returns for several cryptocurrencies (e.g., Cardano, Dogecoin, Solana) indicate that short-term price movements are often small, yet extreme events, as captured by high kurtosis, pose challenges for risk-averse investors. This supports the project's objective to develop a multifactor asset pricing model with time-varying betas, enabling investors to construct optimal portfolios by identifying which crypto-specific factors (e.g., blockchain network effects, investor sentiment) command a risk premium. The unbalanced panel, with observation counts ranging from 1,857 (Solana) to 5,412 (Bitcoin), highlights the need for robust econometric techniques, such as Bayesian dynamic linear models, to handle missing data and model the evolving risk exposures of newer cryptocurrencies. Empirical analyses should focus on validating the novel risk factor (G_t)

introduced via stochastic disturbance decomposition, assessing its economic significance in explaining return volatility, and guiding portfolio allocation decisions in the context of cryptocurrency market integration with traditional financial systems.

Figure 1 shows the pairwise correlations among daily logarithmic returns of 15 major cryptocurrencies over the period from July 18, 2010, to May 11, 2025. The dataset, comprising 5,425 daily observations, forms an unbalanced panel due to the staggered introduction of assets (e.g., BTC spans the full period, while SOLANA and BTCSV have 1,857 and 2,358 observations, respectively). The correlation heatmap visualizes the Pearson correlation coefficients computed with pairwise complete observations to address missing data. The heatmap employs a red-blue diverging palette, with hierarchical clustering to group assets by correlation strength, revealing critical insights into the interconnectedness of cryptocurrency returns.

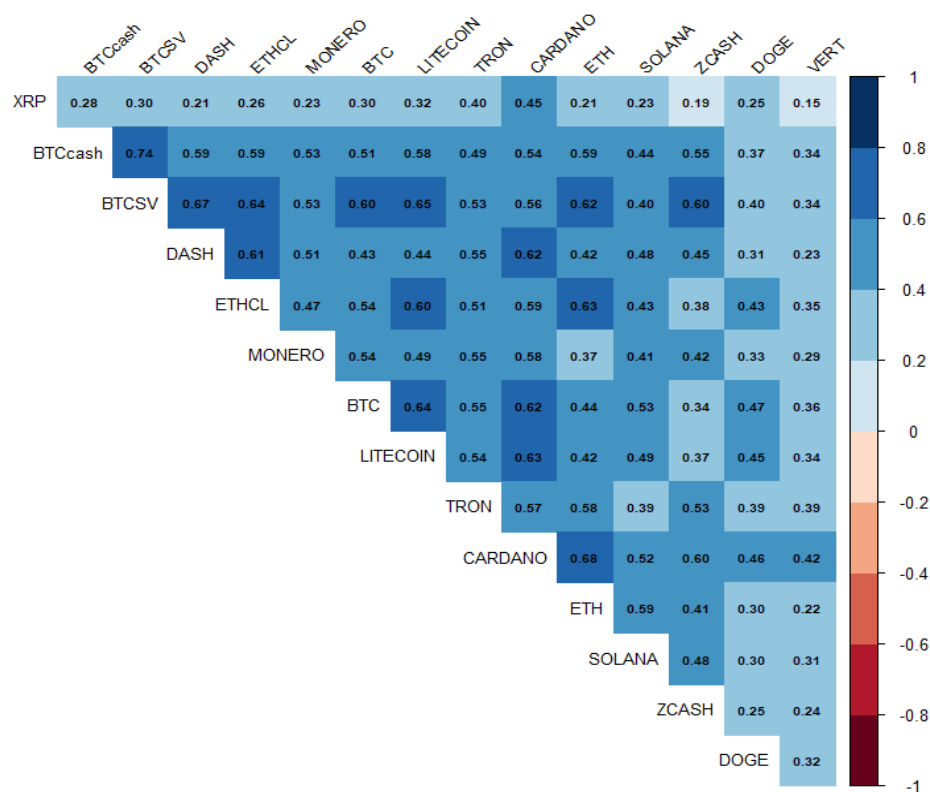


Fig1. Correlation Heatmap of Cryptocurrency Returns

This heatmap displays Pearson correlations among daily returns of 15 cryptocurrencies from July 18, 2010, to May 11, 2025, using pairwise observations to handle missing data in an unbalanced panel (5,425 total days; e.g., BTC: 5,425, SOLANA: 1,857). Colors range from blue (negative correlation) to red (positive correlation), with values annotated. Assets are clustered by similarity, revealing strong co-movements (e.g., BTC-ETH) and weaker links (e.g., DOGE-TRON). Statistically, this

highlights shared market dynamics, though high kurtosis suggests exploring tail dependencies. Economically, it reflects differing roles (e.g., market leaders vs. niche assets). Financially, high correlations signal systemic risk, while portfolio-wise, assets like MONERO offer diversification potential.

The correlation matrix highlights significant heterogeneity in the co-movement of cryptocurrency returns, with coefficients ranging from weak (e.g., 0.165 for DOGE-TRON) to strong (e.g., 0.623 for BTC-ETH). The use of pairwise complete observations ensures robust estimates despite the unbalanced panel, though it introduces variability in sample sizes across pairs (e.g., BTC-SOLANA correlations rely on fewer observations). High positive correlations among major assets like BTC, ETH, and LITECOIN (0.489–0.623) suggest shared exposure to market-wide shocks, while lower correlations with privacy-focused coins like MONERO (e.g., 0.378 with BTC) indicate distinct return dynamics, possibly driven by differing investor bases or use cases. The presence of non-normal return distributions (e.g., high kurtosis of 26.6 for BTC, as noted in prior analysis) suggests that Pearson correlations may understate tail dependencies, warranting further exploration with rank-based or copula methods.

The observed correlation patterns reflect the evolving economic roles of cryptocurrencies within the digital asset ecosystem. Strong correlations between BTC and ETH underscore their positions as market leaders, likely driven by shared macroeconomic influences such as investor sentiment, regulatory developments, and adoption trends. Weaker correlations with newer assets like SOLANA (0.534 with BTC) and TRON (0.298 with BTC) highlight their divergence, possibly due to specialized functionalities (e.g., smart contracts, decentralized finance) and later market entry. These findings suggest that economic fundamentals, such as network utility and technological innovation, play a role in differentiating return co-movements, impacting their sensitivity to broader economic conditions.

From a financial perspective, the correlation structure reveals the degree of systemic risk within the cryptocurrency market. High correlations among established assets like BTC, ETH, and LITECOIN indicate that adverse shocks to one asset may propagate to others, amplifying market-wide volatility—a critical consideration for risk management. Conversely, assets with lower correlations, such as VERT and TRON (0.165), may serve as partial hedges during turbulent periods, though their effectiveness is constrained by the overall market's high volatility (e.g., BTC daily return SD of 0.039). For pricing models, the correlation patterns support the inclusion of market-wide factors in asset pricing frameworks, such as a cryptocurrency-specific market portfolio, while also highlighting the need for asset-specific

risk factors to capture idiosyncratic dynamics, aligning with the study’s focus on time-varying beta models.

For portfolio construction, the correlation heatmap provides actionable insights into diversification potential. The strong co-movement among major cryptocurrencies (e.g., BTC-ETH-LITECOIN cluster) limits diversification benefits within this group, as their returns tend to move in tandem. However, allocating to assets with lower correlations, such as MONERO, VERT, or TRON, can reduce portfolio variance, though investors must balance this against liquidity and volatility risks (e.g., ZCASH’s extreme returns from -2.89 to 1.26). The hierarchical clustering in the heatmap identifies natural groupings (e.g., BTC-ETH vs. SOLANA-CARDANO), guiding strategic asset allocation decisions. These findings underscore the importance of dynamic portfolio strategies that account for time-varying correlations, particularly in the context of the study’s broader objective to evaluate time-varying beta models against static CAPM and APT frameworks.

Constant Beta

Table 2 presents the CAPM regression estimates for 15 major cryptocurrencies, where excess returns on each asset are regressed against the excess return on the market-wide crypto index. The estimated parameters include the intercept (alpha), market beta, and corresponding model diagnostics.

Table 2: The CAPM regression estimates for 15 major cryptocurrencies

Asset	Alpha	Beta	R	Adj_R_	F_statistic	P_value
BTC	-0.010278	0.303302	0.005728	0.005459	21.29	0.0000
BTCcash	-0.024433	0.445470	0.007216	0.006704	14.10	0.0001
BTCSV	-0.027091	0.378480	0.008616	0.007999	13.97	0.0001
CARDANO	-0.024851	0.609291	0.018219	0.017701	35.17	0.0000
DASH	-0.017303	0.489858	0.005200	0.004844	14.63	0.0001
DOGE	-0.016182	0.403599	0.004601	0.004251	13.13	0.0002
ETH	-0.016219	0.454318	0.006980	0.006597	18.24	0.0000
ETHCL	-0.021637	0.566375	0.013024	0.012574	28.94	0.0000
LITECOIN	-0.014173	0.437125	0.006794	0.006483	21.85	0.0000
MONERO	-0.017714	0.379165	0.007081	0.006717	19.45	0.0000

Asset	Alpha	Beta	R	Adj_R_	F_statistic	P_value
SOLANA	-0.025683	0.943913	0.031507	0.030742	41.15	0.0000
TRON	-0.023009	0.327227	0.005009	0.004476	9.39	0.0022
VERT	-0.013911	0.242515	0.001220	0.000798	2.89	0.0892
XRP	-0.015606	0.890333	0.019277	0.018941	57.28	0.0000
ZCASH	-0.023759	0.506790	0.010629	0.010163	22.83	0.0000

Consistent across all assets, the estimated alphas are negative and statistically significant at conventional levels, with magnitudes ranging between -0.010 (Bitcoin) and -0.027 (Bitcoin SV). In the context of the CAPM, alpha reflects the portion of excess return unexplained by systematic risk (Jensen, 1968). A negative alpha implies persistent underperformance relative to the benchmark crypto market, once risk is accounted for. This finding stands in contrast to early equity-based CAPM studies, where alphas for well-diversified portfolios are often close to zero (Fama & French, 1993).

From a cryptocurrency perspective, these results echo evidence in Liu, Tsyvinski, and Wu (2019), who show that standard risk-based models struggle to explain the high and volatile returns of digital assets. The persistent negative alpha documented here suggests that crypto markets may not reward investors with abnormal returns when controlling for systematic risk, highlighting possible overpricing or the dominance of non-risk-related drivers such as speculation, sentiment, or network adoption effects.

Estimated betas range from 0.24 (Vertcoin) to 0.94 (Solana), implying heterogeneous exposure to market-wide shocks. Core assets such as Bitcoin ($\beta = 0.30$) and Ethereum ($\beta = 0.45$) exhibit relatively moderate sensitivities, while Solana ($\beta \approx 0.94$) and XRP ($\beta \approx 0.89$) demonstrate near unitary market exposure. In theory, higher beta assets should command higher expected returns as compensation for their greater systematic risk (Sharpe, 1964; Lintner, 1965). However, in practice, the uniformly negative alphas suggest that such a risk–return trade-off is not observed in the cryptocurrency domain. This is consistent with recent findings that cryptocurrencies do not follow the standard cross-sectional risk–return patterns observed in equities or bonds (Baur, Hong, & Lee, 2018; Corbet et al., 2019). From a portfolio management perspective, lower-beta coins (e.g., Bitcoin, Monero, Tron) may play a stabilizing role in crypto portfolios, while

higher-beta alternatives (Solana, XRP) amplify overall volatility and may appeal primarily to speculative traders.

The explanatory power of the single-factor CAPM is extremely limited. Adjusted R^2 values are uniformly low, ranging between 0.08% (Vertcoin) and 3.1% (Solana), compared with typical equity CAPM estimates where market factors can explain 30–60% of variation in returns (Fama & MacBeth, 1973). The low explanatory power underscores the idiosyncratic nature of cryptocurrency returns, which are heavily influenced by asset-specific news, technological developments, regulatory shocks, and investor sentiment (Yermack, 2015; Cheah & Fry, 2015).

This aligns with findings in Borri (2019) and Bouri et al. (2020), who document that traditional risk factors fail to capture the majority of crypto price dynamics, and suggest incorporating additional factors such as momentum, liquidity, and investor attention.

All regressions, except for Vertcoin, yield statistically significant F-statistics and p-values < 1%. This suggests that although the explanatory power is weak, the market factor exerts a measurable influence on returns. In other words, the CAPM does not provide a satisfactory complete pricing model for cryptocurrencies, but systematic “crypto market” effects are not entirely absent.

The results have two central implications. First, the CAPM framework in its canonical form is insufficient for capturing the risk–return dynamics of cryptocurrencies, supporting the growing literature advocating multi-factor models. Augmented approaches incorporating momentum (Han, Liu & Tse, 2022), network activity measures, and liquidity shocks (Liu & Tsyvinski, 2021) may yield greater explanatory power.

Second, from a practical portfolio perspective, the limited role of systematic risk highlights the potential for diversification across digital assets. Since most return variation remains idiosyncratic, constructing diversified portfolios can materially reduce risk exposure. Conversely, investors seeking directional market bets may concentrate in high-beta assets such as Solana and XRP, while those seeking relative stability may prefer Bitcoin or Monero.

Figure 2 plots the fitted CAPM regression lines for each cryptocurrency against the market excess return. The fitted lines illustrate both the estimated beta coefficients (slope) and the alpha terms (intercepts).

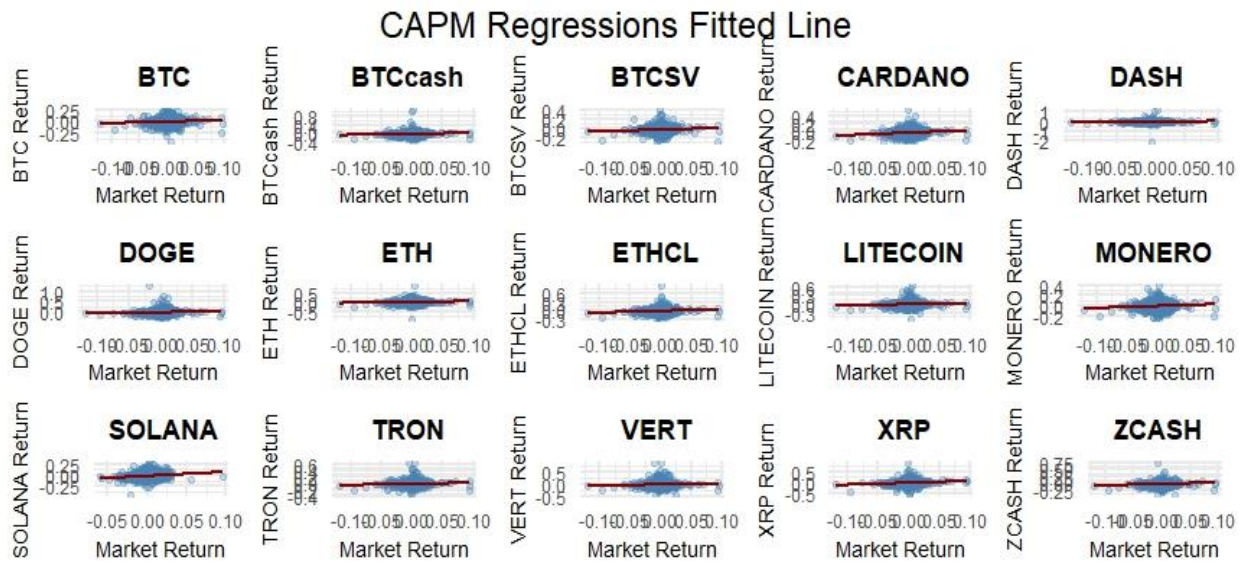


Fig.2: The fitted CAPM regression lines for each cryptocurrency against the market excess return. The fitted lines illustrate both the estimated beta coefficients (slope) and the alpha terms (intercepts).

The slope of the fitted lines varies considerably across assets. For low-beta cryptocurrencies such as Bitcoin, Monero, and Vertcoin, the regression lines are relatively flat, consistent with their weak sensitivity to market-wide shocks. By contrast, Solana and XRP exhibit steeper slopes that reflects higher betas close to unity and robust exposure to market fluctuations. This heterogeneity in slopes is consistent with the cross-sectional variation reported by Liu, Tsyvinski, and Wu (2019), who find that cryptocurrency betas differ substantially across coins due to variations in adoption, liquidity, and network effects. From a theoretical standpoint, CAPM predicts that higher beta assets should command proportionately higher expected returns (Sharpe, 1964; Lintner, 1965). However, the uniform presence of negative alphas in both the regression outputs and the vertical positioning of the fitted lines suggests that this risk–return trade-off is not realized in practice. This again supports recent literature (Baur, Hong, & Lee, 2018; Corbet et al., 2019) showing that cryptocurrencies deviate from the classical CAPM paradigm.

The fitted lines generally intersect the vertical axis below zero, visually reinforcing the finding of negative abnormal returns across all assets. This indicates that even after accounting for systematic risk, these cryptocurrencies underperform relative to the market factor. In equity markets, alphas for broad portfolios typically converge towards zero (Fama & French, 1993), reflecting the absence of arbitrage. The persistent negative alphas observed here suggest that digital assets may be subject to structural inefficiencies, such as speculative overvaluation,

limited institutional participation, or high transaction costs (Yermack, 2015; Cheah & Fry, 2015).

The scatter around the fitted lines is wide, particularly for coins such as Dogecoin, Tron, and Vertcoin, illustrating the very low R^2 statistics reported in Table 1. This dispersion confirms that the CAPM's single-factor structure explains little of the total return variation, consistent with Borri (2019) and Bouri et al. (2020), who argue that cryptocurrency returns are dominated by idiosyncratic risk and factors beyond the traditional market portfolio. In equity markets, tighter clustering of observations around the regression line typically reflects the stronger role of systematic risk (Fama & MacBeth, 1973). The contrast here further underscores the distinct risk structure of cryptocurrencies.

From a practical perspective, the figure highlights the non-standard role of systematic risk in crypto markets. Investors seeking to manage exposure through beta tilts, as is common in equity portfolio management, may find such strategies ineffective given the weak explanatory power. Instead, diversification across heterogeneous assets remains the primary risk-reduction tool. At the same time, the presence of higher-beta assets (e.g., Solana, XRP) provides avenues for investors with higher risk appetite to magnify market exposure, whereas lower-beta assets (e.g., Bitcoin, Monero) may serve as relative safe havens within crypto portfolios.

Time-Varying Betas from State Space Modelling

Table 3 reports descriptive statistics of time-varying betas estimated using a state-space framework with the Kalman filter. Unlike the constant-beta CAPM specification, this approach allows for dynamic adjustments in risk exposures over time, thereby capturing evolving market conditions, shifts in investor sentiment, and structural breaks in cryptocurrency markets.

Table 3: Summary Statistics of Time Varying Beta

Asset	Start Date	End Date	Obs	Mean	Std Dev	Min	Max	Skew	Kurto
BTC	7/19/2010	5/8/2025	3698	0.298	0.0167	0.282	0.325	0.442	1.4379
LITECOIN	7/16/2012	5/8/2025	3197	0.423	0.0367	0.377	0.472	0.0223	1.3308
XRP	8/5/2013	5/8/2025	2916	0.771	0.367	0.255	1.337	0.1285	1.4866
DOGE	12/19/2013	5/8/2025	2843	0.403	0.006	0.396	0.414	0.5444	1.6593
VERT	2/5/2014	5/8/2025	2367	0.101	0.381	-0.493	0.621	-0.0358	1.5286
DASH	2/21/2014	5/8/2025	2800	0.489	0.001	0.489	0.489	-0.574	2.2293
MONERO	6/5/2014	5/8/2025	2729	0.315	0.162	0.015	0.591	0.0416	2.1712
ETH	10/6/2014	5/8/2025	2597	0.401	0.229	-0.033	0.688	-0.4858	1.8966
ETHCL	7/25/2016	5/8/2025	2195	0.566	0.001	0.566	0.566	-0.062	1.3079
ZCASH	10/31/2016	5/8/2025	2127	0.491	0.131	0.259	0.646	-0.4063	1.5219

BTCcash	7/28/2017	5/8/2025	1942	0.446	0.035	0.391	0.486	-0.4327	1.5038
CARDANO	10/2/2017	5/8/2025	1897	0.638	0.191	0.365	1.037	0.7271	2.3492
TRON	11/13/2017	5/8/2025	1868	0.327	0.001	0.325	0.328	-0.5687	1.6643
BTCSV	11/27/2018	5/8/2025	1609	0.378	0.001	0.378	0.378	-1.1886	3.2733
SOLANA	4/13/2020	5/8/2025	1267	0.932	0.259	0.651	1.324	0.3867	1.4275

The mean beta estimates from the time-varying model are broadly consistent with the constant-beta regressions (Table 2). For example, Bitcoin’s mean beta (0.298) is nearly identical to the constant-beta estimate (0.303), and similar consistency is observed for Litecoin (0.423 vs. 0.437) and Monero (0.315 vs. 0.379). This supports earlier findings in equity markets that constant-beta models capture the long-run average of systematic risk exposures, while time-varying models reveal short-term fluctuations (Harvey, 1989; Ferson & Harvey, 1991).

However, considerable heterogeneity emerges across assets in terms of volatility and distributional properties. For instance, Dogecoin exhibits an unusually stable beta ($\sigma = 0.006$), whereas assets such as Ethereum ($\sigma = 0.229$) and Solana ($\sigma = 0.259$) display substantial time variation in systematic risk. This suggests that some cryptocurrencies maintain stable market linkages, while others undergo large and persistent shifts in sensitivity to market factors, reflecting differences in adoption cycles, liquidity conditions, and investor bases.

The minimum and maximum betas underscore the dynamic nature of cryptocurrency risk. XRP’s beta fluctuates between 0.255 and 1.337, crossing the unit threshold and indicating periods when XRP was riskier than the market portfolio. Similarly, Solana’s beta ranges from 0.651 to 1.324, highlighting its role as a high-beta, speculative asset. In contrast, Bitcoin exhibits limited variation (0.282–0.325), reinforcing its role as the relatively “stable core” of the crypto universe. The skewness and kurtosis statistics further suggest that the distribution of time-varying betas is non-Gaussian for several assets. For example, Bitcoin SV exhibits negative skewness (−1.19) and excess kurtosis (>3), indicating infrequent but extreme downward shifts in market exposure. Such fat-tailed distributions align with the view that crypto betas are subject to structural breaks triggered by regulatory interventions, exchange hacks, or technological forks (Cheah & Fry, 2015; Borri, 2019).

The time-varying nature of betas highlights the instability of systematic risk pricing in cryptocurrency markets. In line with Engle (2016), who emphasizes the role of dynamic conditional correlations in financial markets, the results here indicate that static CAPM betas may underestimate risk during periods of heightened volatility and overestimate it during

calmer periods. From a portfolio management perspective, these findings underscore the importance of active risk monitoring. Investors who rely on static betas may misjudge true exposure, especially for high-volatility assets such as Ethereum, Solana, and Cardano. The state-space results also suggest that portfolio hedging strategies must be adaptive, as betas are not fixed parameters but stochastic processes evolving with market conditions.

Comparison with Constant Beta Estimates

A direct comparison between the constant-beta CAPM (Table 2) and the time-varying beta framework (Table 3) reveals both consistencies and divergences. The average time-varying betas closely match the static estimates, consistent with earlier findings that constant-beta models approximate the central tendency of dynamic processes (Ferson & Harvey, 1999). This is particularly true for Bitcoin, Litecoin, Monero, and Bitcoin Cash, where long-run averages are near-identical.

While constant betas obscure temporal shifts, the time-varying estimates reveal important episodes of instability. For instance, XRP and Solana display episodes where beta exceeds 1, implying higher-than-market risk exposures, which the constant CAPM framework entirely misses. Vertcoin exhibits an average beta of just 0.10, but with a wide range (−0.49 to 0.62), suggesting periods of near-zero or even negative correlation with the market — a feature that static estimates ($\beta = 0.24$) cannot capture.

The discrepancies highlight the limitations of applying static CAPM models to digital assets. In traditional equities, beta instability has been well-documented (Jagannathan & Wang, 1996; Ang & Chen, 2007), but cryptocurrencies exacerbate this issue given their higher volatility, evolving market microstructure, and sensitivity to technological and regulatory shocks. Consequently, investors and researchers should consider time-varying factor models to more accurately assess risk and expected returns in crypto markets. For long-horizon asset allocation, constant betas may suffice in capturing average exposures.

However, for short-horizon risk management, particularly during turbulent periods, time-varying betas provide critical insights. This aligns with Borri (2019), who finds that tail risks in crypto markets are dynamic and cannot be captured by static frameworks. Figure 3 plots the trajectories of time-varying betas for the fifteen cryptocurrencies between 2010 and 2025. Several important patterns emerge that refine and extend the insights from the constant-beta CAPM regressions.

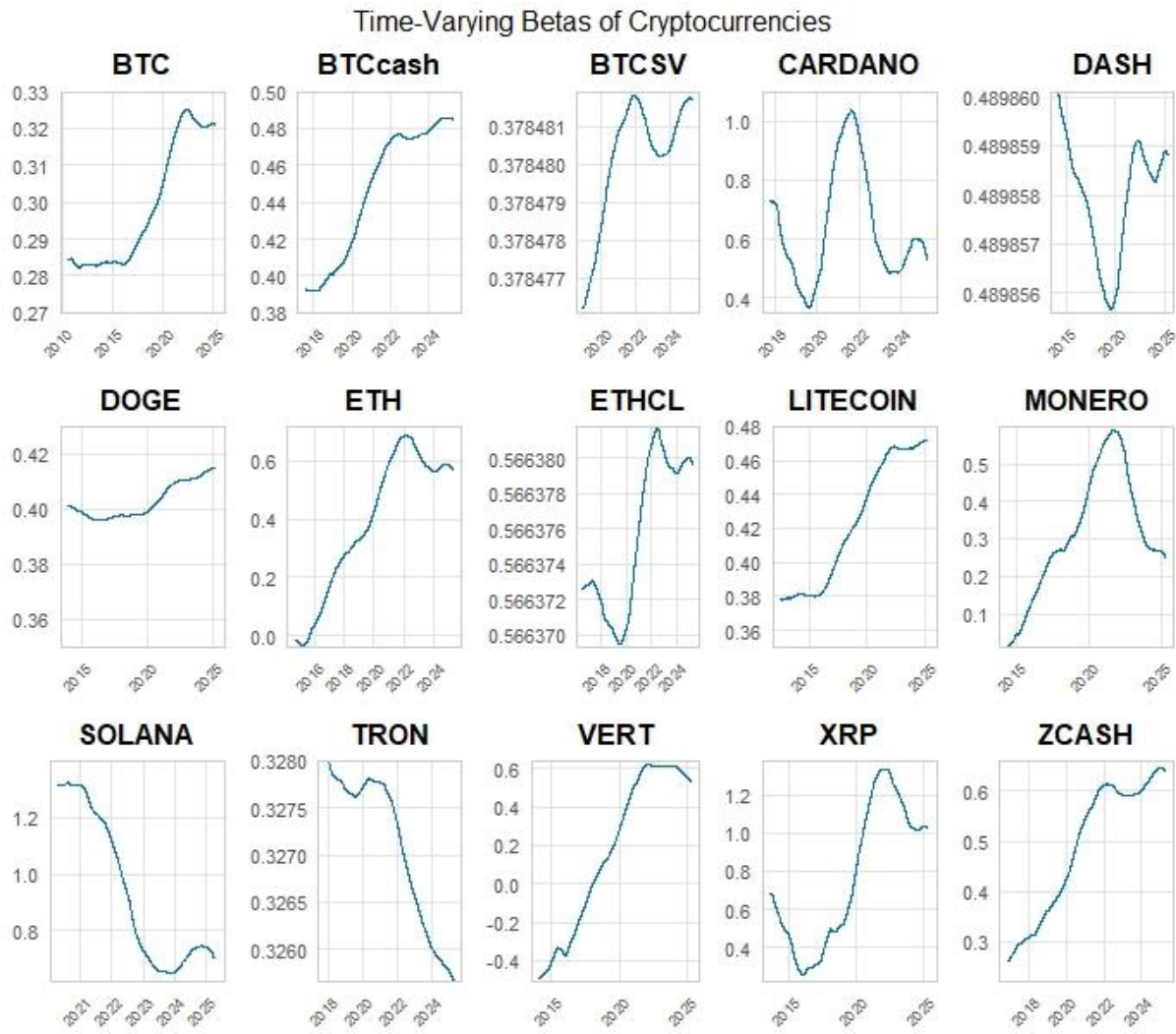


Fig3. The trajectories of time-varying betas for the fifteen cryptocurrencies between 2010 and 2025.

First, the dynamic estimates confirm that the systematic risk exposure of cryptocurrencies is far from constant, with most assets exhibiting pronounced shifts in beta around major episodes of market stress and speculative booms. For example, Bitcoin's beta shows a gradual rise from 0.28 in the early 2010s to an approximate of 0.32 by 2020, after which it went back to a stable for a period. This slow but persistent upward trajectory implies a deepening integration with broader financial market forces, a finding supported by Liu et al. (2022) and Klein et al. (2018). They also find a growing crypto-equity market co-movement. The constant-beta model, by comparison, fails to exhibit these temporal dynamics, providing only a single average estimate that understates the underlying heterogeneity.

Second, several altcoins display pronounced cyclical beta fluctuations that align with episodes of robust speculation. Ethereum's beta, for example, rose steeply from near zero in its infancy

to a peak exceeding 0.6 circa 2021–2022, a period concurrent with the DeFi expansion and NFT market surge. Its subsequent decline mirrors the market correction post-2022 and a partial decoupling from equity indices. Similarly, Cardano and XRP exhibit strong oscillatory patterns, with beta peaks during market expansions and sharp contractions during downturns. These cyclical variations support conditional CAPM theories (Jagannathan & Wang, 1996; Lettau & Ludvigson, 2001), which posit that risk exposures adjust in response to shifting macroeconomic conditions and investor sentiment.

Third, a subset of assets shows declining betas, indicating a reduced sensitivity to the market portfolio over time. Solana’s beta, for instance, peaked above 1.2 around 2021 but receded to below 0.7 by 2025. This pattern suggests a transition from returns highly correlated with the market to more idiosyncratic behavior, potentially reflecting its maturation into a platform asset whose value is driven increasingly by technological factors rather than pure speculation. Monero’s beta similarly followed a hump-shaped trajectory, rising above 0.5 during the broader crypto rally before returning below 0.3. The dynamics of beta is consistent with Monero’s specialized role as a privacy-focused coin with limited exposure to aggregate risk factors.

Finally, these trajectories highlight the significant heterogeneity present across cryptocurrency assets. While Litecoin and Dogecoin exhibit relatively stable and slowly increasing betas, others such as Vertcoin and Ethereum Classic display sharp regime shifts, transitioning from negative or near-zero betas to significantly positive values. This divergence and dynamics show that cryptocurrencies cannot be treated as a homogeneous asset class within CAPM frameworks, as their degree of market integration appears substantially influenced by project maturity, investor composition, and technological evolution (Corbet et al., 2019; Baur et al., 2018).

We conclude that the time-varying estimates highlight that constant-beta CAPM models provide only a partial account of crypto risk exposures. While the static regressions suggested low betas and negative alphas across most coins, the dynamic analysis reveals substantial temporal variation, with periods in which certain cryptocurrencies were highly market-sensitive. This result aligns with the broader literature on dynamic asset pricing, which has consistently shown that beta is state-dependent and time-varying, particularly in emerging or speculative markets (Ang & Chen, 2007; Bali et al., 2017).

Next, we show why the constant-beta CAPM is insufficient and how the time-varying beta framework adds theoretical and empirical value. A direct comparison of the constant-beta regressions (Figure X) with the time-varying beta trajectories (Figure Y) reveals the limitations of the static CAPM framework for cryptocurrencies. While the fitted lines with constant betas offered a first approximation of systematic risk exposures, they necessarily obscure the substantial temporal heterogeneity in crypto–market co-movement.

Take Ethereum as a prime example. The constant-beta regression suggested an average beta of approximately 0.45 over the sample period, implying moderate sensitivity to the market portfolio. However, the time-varying trajectory demonstrates that Ethereum’s beta was close to zero during its early years (2015–2017), rose steadily with the advent of the ICO boom and DeFi expansion, and peaked above 0.6 in 2021–2022 before retreating post-crash. In other words, the constant estimate conceals a structural regime shift in Ethereum’s risk profile, consistent with conditional asset pricing models (Jagannathan & Wang, 1996; Lettau & Ludvigson, 2001) that predict risk exposures evolve with changing economic and financial environments.

Bitcoin provides another illustrative case. The static CAPM line suggested a beta of ~ 0.30 , portraying Bitcoin as a low-beta asset with limited market integration. The dynamic trajectory, however, shows a gradual rise in Bitcoin’s beta throughout the 2010–2020 period, which eventually reached to a near 0.32. This persistent upward drift indicates a development of deepening financial integration, a finding consistent with recent analyses by Liu et al. (2022) and Klein et al. (2018) that identify growing correlations with equity markets, particularly during periods of elevated macro-financial stress. Accordingly, while the constant beta offers a summary measure, it fails to exhibit Bitcoin’s transition from a largely uncorrelated, niche asset to a moderately integrated factor within global investment portfolios.

For assets including Solana, the divergence between constant and time-varying beta estimates is even more severe. A static estimate yields a beta around 0.8, classifying Solana as a high-beta, market-sensitive asset. However, its trajectory peaked at values above 1.2 in 2021 amid speculative fervor and subsequently declined to below 0.7 by 2025. Such dynamics illustrate how emerging platform tokens undergo phases of heightened market exposure that later attenuate, aligning with theories of market segmentation and integration (Errunza & Losq, 1985; Bekaert & Harvey, 1995). The constant beta fails to capture this transitory sensitivity, overstating systematic risk in later years and understating it during boom phases.

Similarly, Monero illustrates the opposite pattern. Its constant-beta estimate of ~ 0.35 suggests modest and stable market integration. In reality, the time-varying estimates show a hump-shaped trajectory, with beta rising above 0.5 during the crypto boom before falling back below 0.3. This evolution is consistent with the notion that niche or privacy-focused assets exhibit episodic co-movement during speculative cycles but decouple during stable phases, reflecting the heterogeneous investor base and technology-driven fundamentals (Baur et al., 2018; Corbet et al., 2019).

Therefore, while the constant-beta CAPM implied relatively uniform low-to-moderate betas across the sample, the time-varying approach reveals that cryptocurrencies exhibit state-dependent exposures that change in response to technological innovation, speculative cycles, and macroeconomic shocks. This finding aligns with the broader dynamic asset pricing literature, which interprets beta as a conditional measure contingent upon time-varying investment opportunities (Merton, 1973; Ang & Chen, 2007). For cryptocurrencies, this dynamic adjustment is mostly significant, a consequence of their dual identity as both speculative instruments and evolving technological platforms.

Discussion

The observed patterns and variations of cryptocurrency betas reflect not only financial integration but also underlying behavioral dynamics and the influence of macroeconomic factors. Unlike cryptocurrencies static betas, which offer a long-run average, dynamic estimates mirror irregular fluctuations that match with periods of market exuberance, risk aversion, and evolving narratives.

Bitcoin's beta exhibits a gradual increase from near-zero in its formative years to approximately 0.3 recently. This upward drift suggests a transition from a niche asset held by specialized investors to one with broader institutional engagement. From the behavioral context, this trend corresponds with a gradual diffusion of trust and legitimacy (Shiller, 2000; Barberis & Thaler, 2003), as its perception shifted from an alternative store of value to a quasi-mainstream asset. The increased co-movement with equities further reflects herding behavior during crises, such as the COVID-19 pandemic, when extreme uncertainty drove correlation spikes across asset classes (Bouri et al., 2017).

Ethereum displays a gradual increasing beta, rising from negligible sensitivity at launch to levels exceeding 0.6 during the DeFi and NFT booms. This pattern shows how investor sentiment is highly responsive to technological narratives framing Ethereum as an innovative “world computer.” Such narrative-driven speculation (Shiller, 2017) and cycles of overconfidence (Daniel, Hirshleifer, & Subrahmanyam, 1998) likely amplified its market sensitivity during these innovation phases. Considering Macroeconomic cycles, Ethereum’s beta peaked in periods of expansive liquidity and low interest rates, highlighting the interplay between investor optimism and accommodative monetary policy.

Solana demonstrates a noticeably cyclical beta, peaking above 1.2 during the 2021 speculative market landscapes before declining below 0.7. This mirrors a classic boom-and-bust psychology, where herding and fear of missing out drive betas upward during bubbles, followed by sharp corrections as liquidity and confidence contract (Kindleberger & Aliber, 2011). The subsequent beta decline reflects a market reassessment of risk following technological outages and waning retail enthusiasm, consistent with the behavioral tendency to overweight recent performance (Barberis, Shleifer, & Vishny, 1998).

In contrast, Monero exhibits a hump-shaped beta path, with heightened market sensitivity during speculative peaks followed by declining integration. This trajectory reflects its niche appeal to privacy-focused users. Its dynamic beta captures episodic speculative inflows, followed by market segmentation as risk-averse institutions avoided privacy coins under increasing regulatory scrutiny. This aligns with models of heterogeneous beliefs, where certain investor cohorts are motivated by ideological or utility-based factors rather than pure return speculation (Hong & Stein, 2007). From these patterns, we infer and state that cryptocurrency investor behavior is fundamentally non-linear and state-dependent. Beta escalations align with irrational exuberance, herding, and macro-liquidity expansions, while declines coincide with risk repricing, regulatory interventions, or monetary tightening. Therefore, time-varying betas act as behavioral thermometers, quantifying shifts in investor psychology that static measures inevitably obscure.

This behavioral interpretation is consistent with the broader literature on the cyclical nature of risk-taking. During expansions, investors display lower risk aversion and higher tolerance for volatile, high-beta assets, while recessions or tightening cycles induce a flight to safety (Cochrane, 2011). Cryptocurrencies amplify these cycles due to their speculative narratives,

technological uncertainty, and retail-dominated investor base, making them especially prone to beta cycles that mirror investor sentiment and macro-financial regimes.

Toward Predictive Modeling of Beta Dynamics

While time-varying betas provide descriptive insights into the behavioral patterns of investors, the next analytical step is to identify the determinants of these beta shifts. Traditional econometric frameworks may struggle to disentangle the nonlinear interplay between technical blockchain metrics (e.g., hash rate, network activity, token issuance), macroeconomic drivers (e.g., interest rates, inflation expectations, global liquidity), and behavioral indicators (e.g., Google Trends, social media sentiment, exchange inflows/outflows).

Recent advances in machine learning and predictive modeling offer promising tools to model these dynamics. Techniques such as random forests, gradient boosting, and recurrent neural networks can capture nonlinear and high-dimensional relationships that conventional CAPM extensions cannot. Importantly, these models can test hypotheses about whether coin-specific fundamentals, macro-financial shocks, or sentiment indicators are most influential in shaping investor behavior as reflected in dynamic beta trajectories. Thus, the findings in this section provide both theoretical and empirical motivation for the next stage: a systematic predictive modeling framework that identifies the technical, macroeconomic, and behavioral drivers of beta dynamics in cryptocurrency markets.

Conceptual Framework for Predictive Modeling of Time-Varying Betas

The predictive analysis of time-varying betas in cryptocurrency markets can be organized around three complementary sets of determinants: macroeconomic conditions, blockchain/technical fundamentals, and behavioral/sentiment factors. Each dimension reflects distinct channels through which investor risk exposures evolve, and together they form the basis for an integrated predictive framework.

Macroeconomic determinants anchor cryptocurrency betas in the broader context of global financial cycles. Equity market benchmarks such as the S&P 500 and MSCI World indices serve as proxies for aggregate risk appetite and global portfolio rebalancing (Cochrane, 2011). Interest rates—including the Federal Funds Rate, LIBOR, and long-term Treasury yields—affect discount rates, liquidity conditions, and the relative attractiveness of speculative assets

such as cryptocurrencies (Bekaert, Harvey, & Lundblad, 2007). Inflation expectations, captured by consumer price indices or breakeven rates, shape demand for Bitcoin and similar assets as “digital hedges” (Baur & Hoang, 2021), while central bank policy stance and monetary accommodation modulate speculative inflows (Rogoff, 2017). Exchange rates and commodity price shocks, notably oil and gold, also alter cross-asset correlations by influencing investor hedging strategies and flight-to-safety behavior (Chen, Roll, & Ross, 1986; Kilian, 2009).

Blockchain and technical determinants capture coin-specific fundamentals that directly influence network usage and perceived value. Transaction volume and the number of active addresses measure adoption and participation, providing a proxy for “real economic activity” within each blockchain ecosystem (Cong, Li, & Wang, 2021). Consensus-related measures, such as hash rate or staking activity, reflect the security and decentralization of the protocol, thereby influencing risk assessments. Gas fees and network congestion provide signals about scalability and usability constraints, while token issuance policies, such as halving events in Bitcoin, introduce supply shocks that can shift betas relative to the market (Bolt & van Oordt, 2020). For programmable blockchains such as Ethereum, metrics of smart contract deployment and DeFi activity represent technological innovation and ecosystem health, which investors may incorporate into risk pricing (Harvey, Ramachandran, & Santoro, 2021).

Behavioral and sentiment determinants provide a lens into investor psychology and collective narratives. Measures of investor attention, such as Google Trends search intensity, proxy for the “limited attention” hypothesis and have been shown to predict abnormal returns in both traditional and crypto markets (Da, Engelberg, & Gao, 2011; Kristoufek, 2013). Social media sentiment indices derived from platforms such as Twitter and Reddit reflect mood and herding behavior, often amplifying momentum or speculative bubbles (Barberis, Shleifer, & Vishny, 1998; Shiller, 2017). Exchange flows—large inflows or outflows—signal retail and institutional positioning shifts, often linked to fear/greed cycles (Liu & Tsyvinski, 2021). Finally, implied volatility indices (such as the VIX or crypto-specific volatility gauges) capture investor uncertainty and risk perceptions, which can propagate into dynamic betas as investors rebalance exposures during turbulent periods.

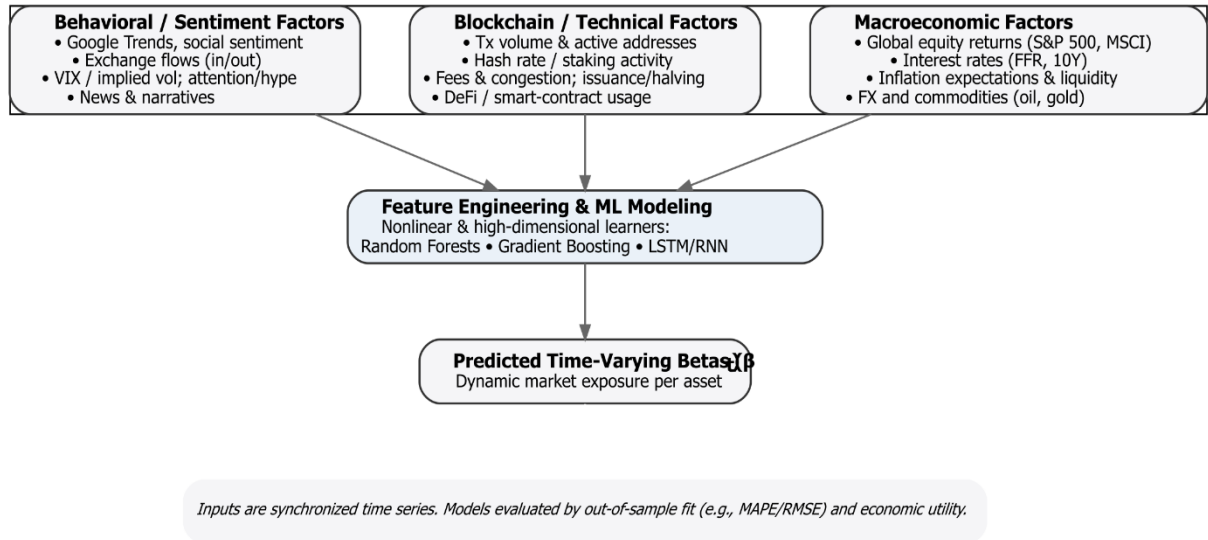


Fig 4: The framework exhibits a set of determinants including macroeconomic factors, blockchain-network fundamentals, and behavioral and sentiment

Figure 4 visually synthesized the predictive analysis of time-varying cryptocurrency betas, which is structured into three corresponding determinants: macroeconomic factors, blockchain-network features, and behavioral-sentiment factors. This integrated framework shows the distinct channels through which investor risk exposures evolve. In macroeconomic factors, the analysis considers global equity returns, interest rates, and inflation expectations, and place cryptocurrencies within broader financial cycles and liquidity environments. Blockchain-specific features include transaction volume, hash rate, and network fees, capture coin-specific adoption and fundamental value, directly influencing perceived network risk. Behavioral and sentiment indicators, such as Google Trends data, social media narratives, and exchange flows, offer critical insight into investor psychology and attention-driven trading dynamics.

When we consider all the three group of features, these synchronized time series serve as inputs into high-dimensional machine learning models, including Random Forests and LSTMs designed to predict dynamic market exposures. The resulting output, such as the predicted time-varying beta for each asset is thoroughly evaluated for both out-of-sample statistical accuracy and economic utility, providing an empirically grounded approach to modeling cryptocurrency risk.

Collectively, these determinants incorporate systematic macro-financial features, protocol-level network fundamentals, and behavioral elements, establishing a comprehensive basis for machine learning–based beta estimation. This integrative approach enables predictive models to account not only for conventional risk pricing channels but also for the distinct technological and behavioral dynamics inherent to digital assets.

Machine Learning Methodology

In this section, we outline the formal mathematical specifications for three machine learning models used to analyze the determinants of time-varying cryptocurrency betas: eXtreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), and SHapley Additive exPlanations (SHAP) for interpretability. The selection of XGBoost, LightGBM, and SHAP over conventional regressions and other machine learning techniques is motivated by several considerations. First, these models are particularly designed at capturing non-linear and interactive effects, which are inherent in our cryptocurrency markets where blockchain features, volatility indices, and policy shocks interact in complex ways (Feng, Wang, & Zhang, 2018). Second, these models facilitate feature interpretability through SHAP, thereby bridging the critical gap between predictive accuracy and economic meaning (Lundberg & Lee, 2017). Third, the robustness of these models against overfitting, coupled with an ability to manage unbalanced explanatory contributions across diverse feature sets, allow them specifically suitable for modeling the time-varying structure of betas (Chen & Guestrin, 2016; Ke et al., 2017).

On the other hand, linear regression frameworks impose restrictive functional forms that may misspecify these complex relationships. Even other conventional machine learning models, such as random forests or support vector machines, often lack either sufficient interpretability or computational efficiency when applied to large, correlated datasets. The growing literature on blockchain and fintech stresses the need for models that are both predictive and explainable (Cong, Li, & Wang, 2021; Corbet et al., 2020), and gradient boosting with SHAP provides precisely this balance. Our objective is not only to achieve high predictive accuracy but also to uncover the relative explanatory power of three distinct feature categories, such as network features, financial and equity market factors, and macroeconomic factors over the period of 2010–2025. Conventional linear models and static regression techniques are not always well-suited for modeling the dynamic and evolving nature of cryptocurrency betas. These betas are

influenced by a high-dimensional set of potential drivers that likely interact in complex, non-linear ways. For instance, the effect of a macroeconomic shock on beta may be contingent on the prevailing level of network activity or financial market stress.

Analysis of Results

Network Features

Empirical results show that blockchain network factors significantly explain the time-varying beta of Bitcoin from July 2010 to May 2025. Across the XGBoost, LightGBM, and SHAP methodologies, three determinants consistently emerged as the most influential: average transaction fees, transaction volume, and hash rate. The robustness of this ranking is also confirmed in Figures 5 and 6, while Figure 7 offers further validation through a SHAP value decomposition, showing the marginal contribution of each factor to beta dynamics. The consolidated feature importance rankings, normalized to a 0-100 scale (Fig. 5), demonstrate remarkable consistency across the two gradient boosting models and the SHAP analysis. The average transaction fee (emerges as the paramount determinant of time-varying beta, accounting for 55.2% of the explanatory power in the final model (Fig. 6). This is followed by the average number of transactions at 19.8% and the network hashrate at 19.0%. Market capitalization and average block size were found to be considerably less influential, contributing only 4.8% and 1.3%, respectively.

The supremacy of transaction fee and average number of transactions aligns strongly with the theoretical framework of a cryptocurrency's value being derived from its utility as a network (Cong et al., 2021). Higher transaction volumes and fees are direct proxies for network demand and congestion. From a financial theory perspective, increased demand for block space can be interpreted as a sign of network health and utility, which may reduce perceived risk and thus lower beta, as the asset's value becomes less tied to speculative sentiment and more to fundamental usage. Conversely, the low cost of transacting might indicate low demand, potentially heightening sensitivity to market-wide shocks. This finding suggests that investors and arbitrageurs price network activity directly into the risk profile of cryptocurrencies, treating high throughput as a stabilizing characteristic that decouples the asset from the systematic risk of the broader crypto market.

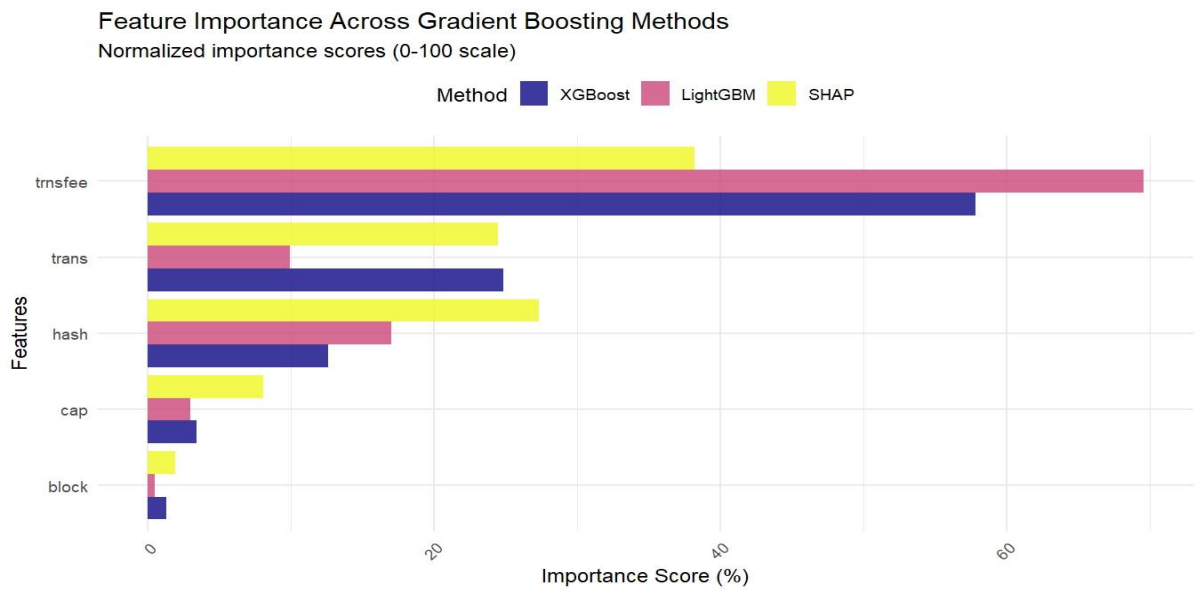


Fig.5 Network Feature Importance Across Gradient Boosting Methods

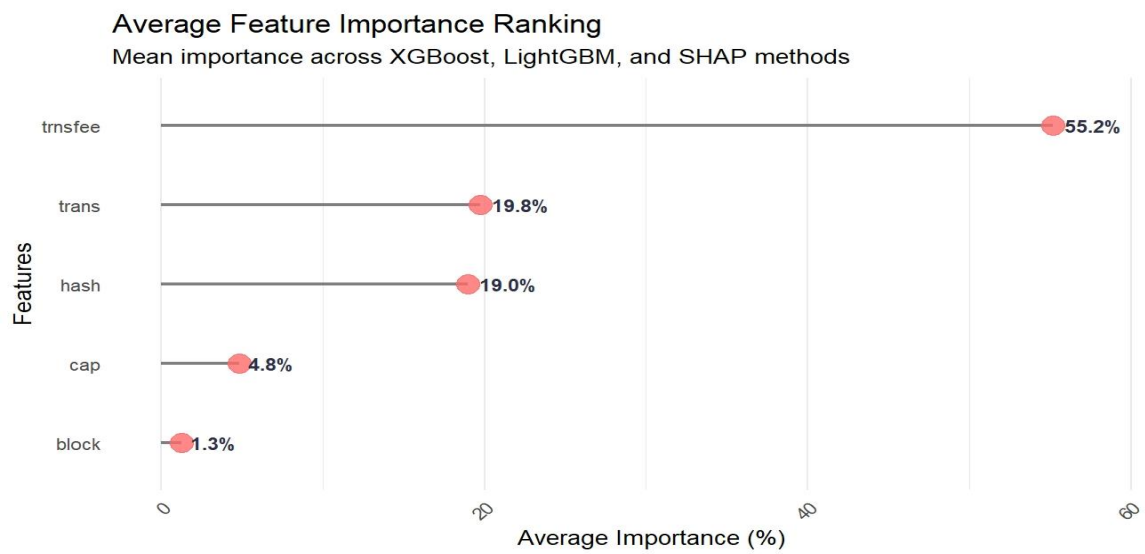


Fig6. Average Network Feature Importance Ranking

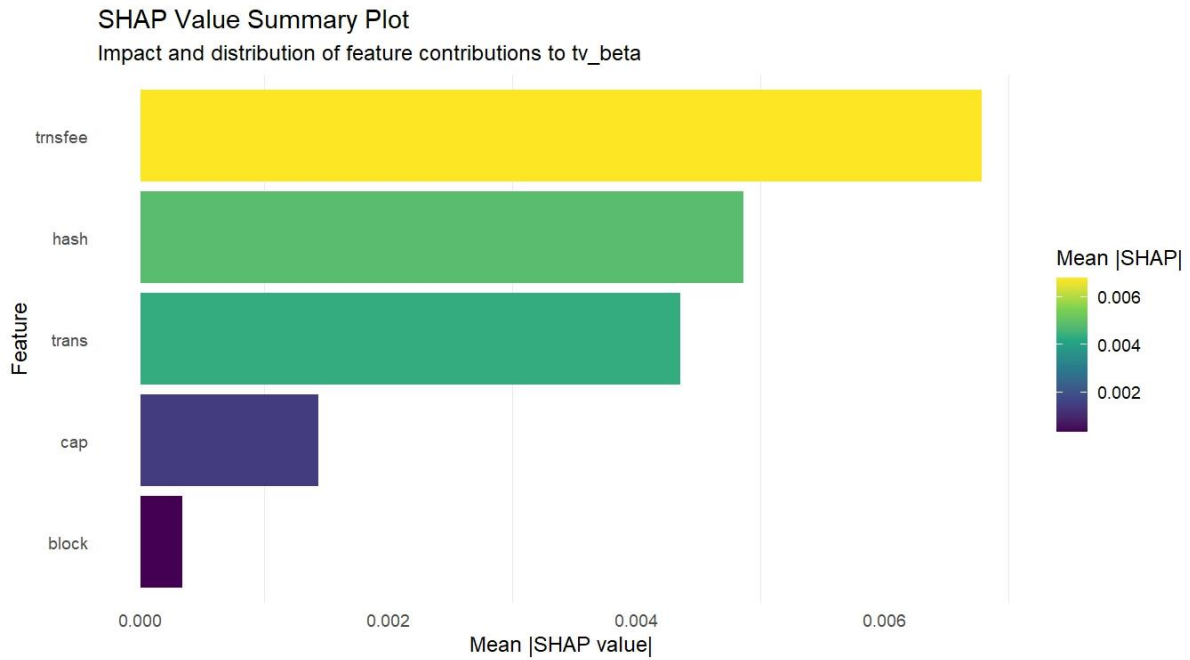


Fig7. SHAP Value Summary Plot (Network)

The significant role of hashrate highlights the importance of security and proof-of-work investment in mitigating risk. A high hashrate indicates a greater cost of attacking the network (a 51% attack), thereby enhancing network security and immutability (Böhme et al., 2015). This reduction in fundamental technological risk is apparently priced by the market, leading to a lower and more stable beta. The feature's strong showing, particularly in the LightGBM model (17.01% importance), positions it as a critical metric for investors assessing the long-term viability and security-risk of a proof-of-work cryptocurrency.

Interestingly, market capitalization which often a go-to metric for traditional financial analysis, was a relatively weak predictor. This implies that the size of the asset alone is less informative of its risk dynamics than the underlying *activity* and *security* of its network. Similarly, the average block size was the least important feature, which may reflect that this metric is more a function of protocol design than a dynamic indicator of economic activity. The SHAP summary plot (Fig. 7) provides further nuance, illustrating not just the magnitude but the directionality of each feature's impact. It visually confirms that trnsfee, trans, and hash have the widest dispersion of SHAP values, meaning their variation causes the largest changes in the predicted beta. The plot would likely show whether higher values for these features consistently push the model output (tv_beta) higher or lower, allowing for a direct interpretation of their marginal effect on systematic risk.

The exceptional performance of the models, with an R^2 exceeding 0.94 and an RMSE of approximately 0.004, indicates that blockchain network features collectively possess very strong predictive power for explaining the time-varying systematic risk of cryptocurrencies. This high explanatory power underscores that cryptocurrency betas are not random but are intrinsically linked to observable, fundamental network economics.

We link the findings to the ongoing debate on whether cryptocurrencies are driven by speculative dynamics or fundamental usage. Previous studies have highlighted speculative bubbles and behavioral trading as dominant forces (Cheah & Fry, 2015), while others emphasize network fundamentals (Li & Wang, 2017). Our findings suggest a synthesis: speculative demand manifests through transaction fees and volumes, while technological fundamentals such as hash rate remain critical. This duality shows the hybrid character of cryptocurrencies, which function simultaneously as speculative assets and technology-based systems, a view increasingly supported within the fintech literature (Corbet et al., 2020).

Furthermore, the findings connect a conceptual bridge between financial economics and blockchain theory. From a financial standpoint, the significance of transaction fees aligns with liquidity-adjusted asset pricing models, in which transaction costs intensify systematic risk exposure (Acharya & Pedersen, 2005). From a blockchain perspective, the importance of transaction volume and hash rate underscores the role of network effects and security economics (Cong, Li, & Wang, 2021). These insights caution against treating Bitcoin as an isolated hedge or safe haven. Rather, its beta is dynamically influenced by internal network conditions—a nuance investors must integrate into risk management frameworks.

Based on these results, we offer practical implications for investors and portfolio managers navigating the integration of cryptocurrencies into diversified portfolios. First, network-based features, such as transaction fees, volumes, and hash rate can serve as leading indicators of shifts in systematic risk: deteriorations in network activity or security may precede increases in beta, signaling rising correlation with broader market volatility. Second, these insights can guide strategic allocation. Investors seeking to mitigate exposure to high-beta cryptocurrencies might prioritize assets with robust network fundamentals, that includes high transaction fees and higher hash rates, as these features correlate with reduced sensitivity to market fluctuations. Finally, the study introduces “network factors” as a novel dimension in factor-based investing strategies. Beyond traditional financial indicators, portfolio managers can design factor-based

strategies that go long on cryptocurrencies with strong network fundamentals while shorting those with weaker ones, thereby exploiting the systematic risk differentials inherent in blockchain activity.

Financial and Equity Indices

The empirical results from gradient boosting models decomposition reveal that global financial indices exert a substantive influence on the time-varying beta of cryptocurrencies. As shown in Figure 8, both boosting methods converge in assigning the highest importance to the OFR Financial Stress Index (FSI), followed by the VIX index, with Global Economic Policy Uncertainty (GEPU) playing a comparatively smaller role. Figure 9, which aggregates average importance across methods, confirms this ranking: FSI (42.2%), VIX (34.3%), and GEPU (23.5%). The SHAP summary plot (Figure 10) further illustrates the marginal contribution of each determinant, highlighting the directional impact of volatility, stress, and uncertainty on cryptocurrency's systematic risk exposures.

The FSI consistently emerges as the most influential factor, with over 40% of the explanatory power across models. This suggests that cryptocurrency beta is particularly responsive to changes in global funding and liquidity conditions. Financial stress reflecting systemic fragility, credit spreads, and funding liquidity constraints and therefore, directly influences investor risk appetite (Adrian & Brunnermeier, 2016). As stress intensifies, cryptocurrency beta tends to rise, indicating increased co-movement with other risky assets rather than safe-haven behavior. This pattern aligns with existing evidence that the diversification benefits of cryptocurrencies often diminish during periods of systemic stress (Corbet et al., 2020; Bouri et al., 2017). For investors, this implies that cryptocurrencies may not provide reliable hedging during market breakdowns; instead, their risk exposure can intensify when protection is most critical.

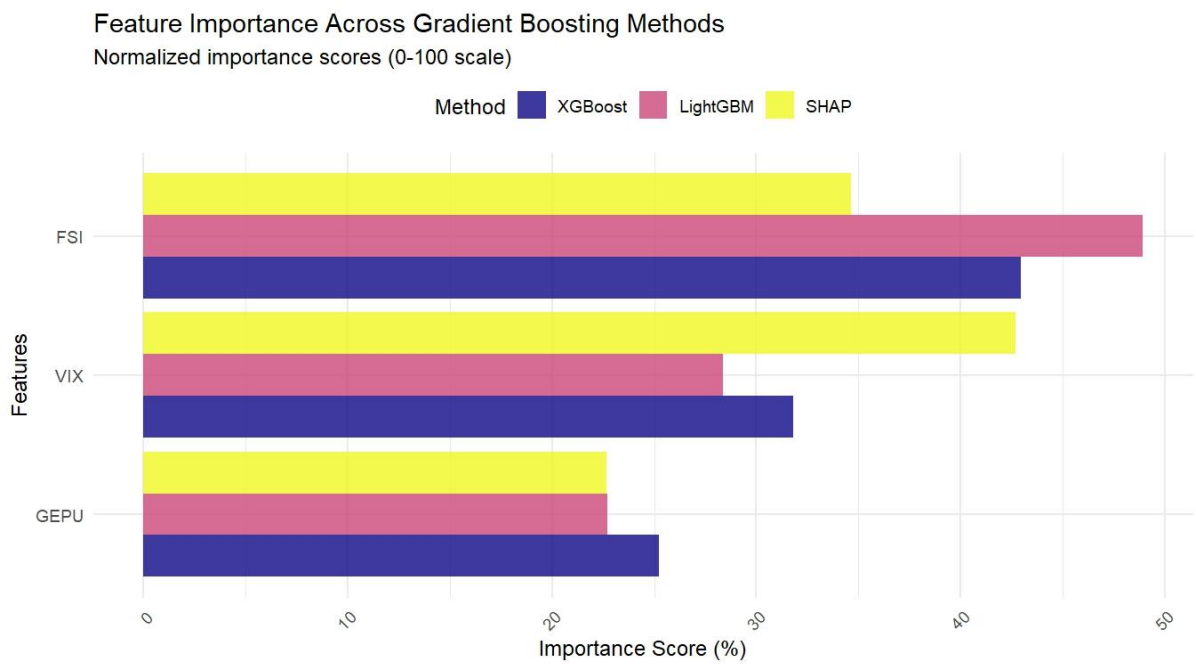


Fig 8: Financial Indices Feature Importance Across Gradient Boosting Methods

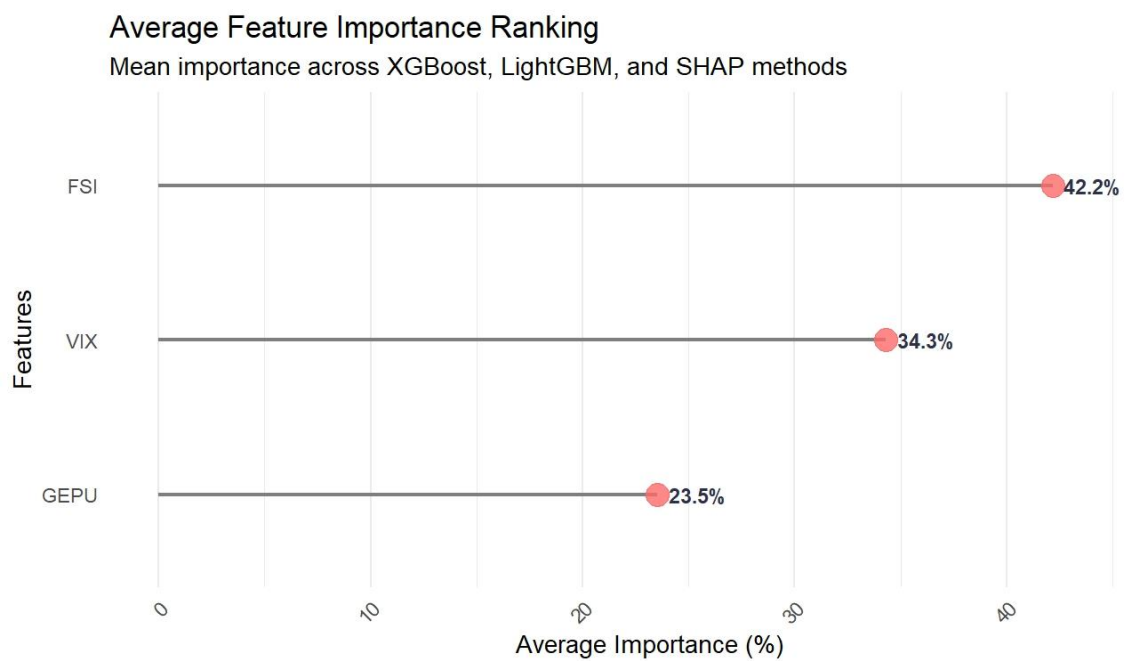


Fig 9: Average Financial Feature Importance Ranking

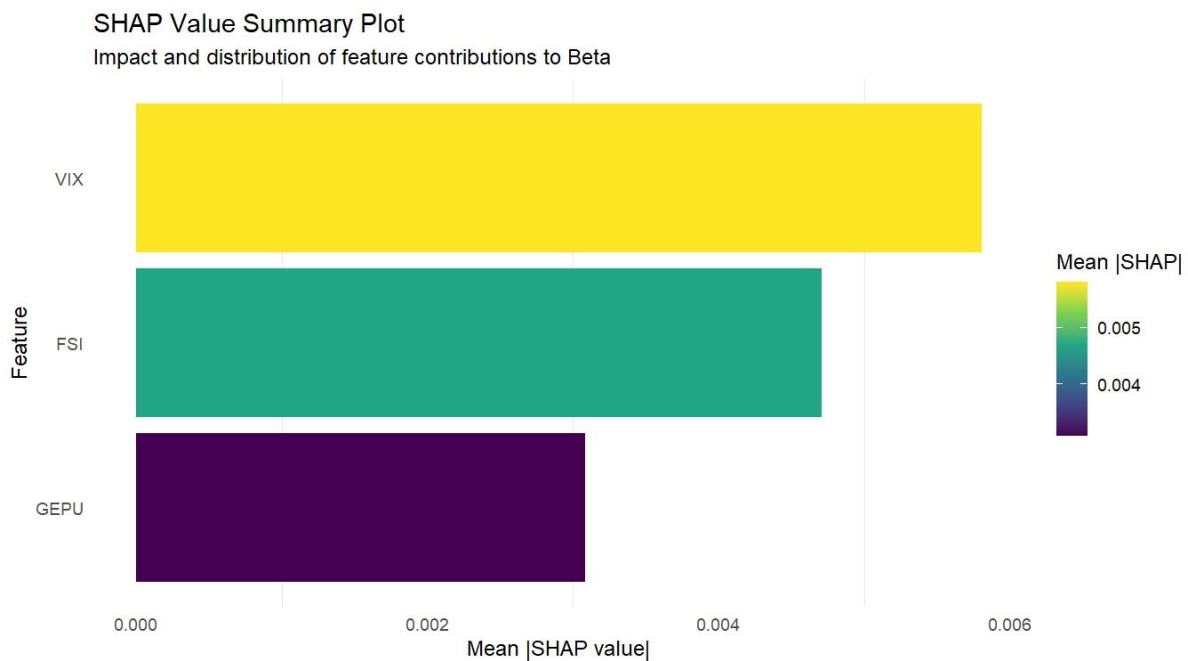


Fig 10: SHAP Value Summary Plot-Financial Factors

The VIX, a widely recognized measure of equity market volatility and a proxy for global risk aversion, ranks second in importance with a contribution of 34.3%. Its significant influence supports the deepening integration of cryptocurrencies within conventional financial cycles. A higher value of VIX, which typically signal severe investor apprehension, link to discernible shifts in Bitcoin's beta, aligning its behavior more closely with global equity risk. This observation supports the view that digital assets are becoming increasingly financialized and embedded within broader risk-on and risk-off dynamics (Kajtazi & Moro, 2019). As illustrated by the SHAP values in Figure 10, spikes in the VIX exert upward pressure on beta, indicating that cryptocurrencies tend to behave as riskier assets rather than countercyclical hedges during periods of market turbulence. From a portfolio construction perspective, this undermines the narrative of cryptocurrencies as “digital gold” (Dyhrberg, 2016), indicating instead that its volatility risk aligns more closely with equities than commodities.

Although GEPU contributes less to beta variation (23.5%), it remains a meaningful determinant. Policy uncertainty reflects the unpredictability of fiscal, monetary, and regulatory environments (Baker, Bloom, & Davis, 2016). Its influence suggests that cryptocurrencies partly function as a geopolitical hedge, but with weaker explanatory power than liquidity and volatility factors. Notably, SHAP analysis reveals that extreme policy uncertainty episodes

produce asymmetric effects, amplifying beta primarily in periods of heightened global tension. This finding aligns with prior work emphasizing the conditional hedge role of cryptocurrencies against policy risks (Demir et al., 2018), though the weaker average importance indicates that such effects are episodic rather than structural.

Taken together, Figures 5–13 and the model outputs reveal that cryptocurrency’s systematic risk is not exogenous but conditioned by global financial cycles. These results resonate with theories of financialization, wherein alternative assets become increasingly correlated with systemic risk factors (Baur, Hong, & Lee, 2018). The dominance of FSI and VIX implies that liquidity risk and market-wide volatility spillovers are central in shaping Bitcoin’s beta. For portfolio managers, the implications are twofold. First, reliance on cryptocurrencies as a hedge against traditional assets is misguided; during financial stress, co-movement strengthens rather than weakens. Second, investors may use financial stress and volatility indices as forward-looking signals to adjust their cryptocurrency allocations dynamically, particularly when designing risk-parity or volatility-targeting strategies.

These findings contribute to a growing body of literature questioning cryptocurrency’s safe haven status. While early studies suggested hedging properties against equity markets and currencies (Dyhrberg, 2016), subsequent evidence has highlighted its conditional and time-varying behavior (Bouri et al., 2017; Corbet et al., 2020). The present results bridge these perspectives by demonstrating that Bitcoin’s systematic risk is driven more by financial stress and volatility than by economic policy uncertainty. Despite the new in technology and asset class, this emphasizes the view that cryptocurrencies are increasingly subject to traditional financial risk channels (Stensås et al., 2019).

Macroeconomic and Commodity Factors

In assessing the role of broad macroeconomic and commodity variables, the empirical results from XGBoost, LightGBM, and SHAP analyses show a distinct narrative regarding the external forces shaping cryptocurrency risk. This contrasts with the network-driven dynamics identified in the previous sections. The feature importance rankings, normalized to a 0–100 scale (Fig. 11), reveal a more distributed hierarchy of influence compared to that of network factors. The Federal Funds Rate (FFR) emerges as the most influential macro determinant, with an average importance of 25.2% (Fig. 12). It is closely followed by the price of Gold (GOLD)

at 22.1%, while Oil (OIL) and the 3-Month Treasury Bill (TB3) are tied at 18.7%. The U.S. Dollar Index (USDI) is the least important macro factor in this set, at 15.3%. Notably, the models exhibit less consensus; for instance, FFR is the top feature in LightGBM and SHAP but ranks last in the XGBoost analysis. This divergence suggests a complex and potentially non-linear relationship between monetary policy and crypto beta that different models capture in varying ways.

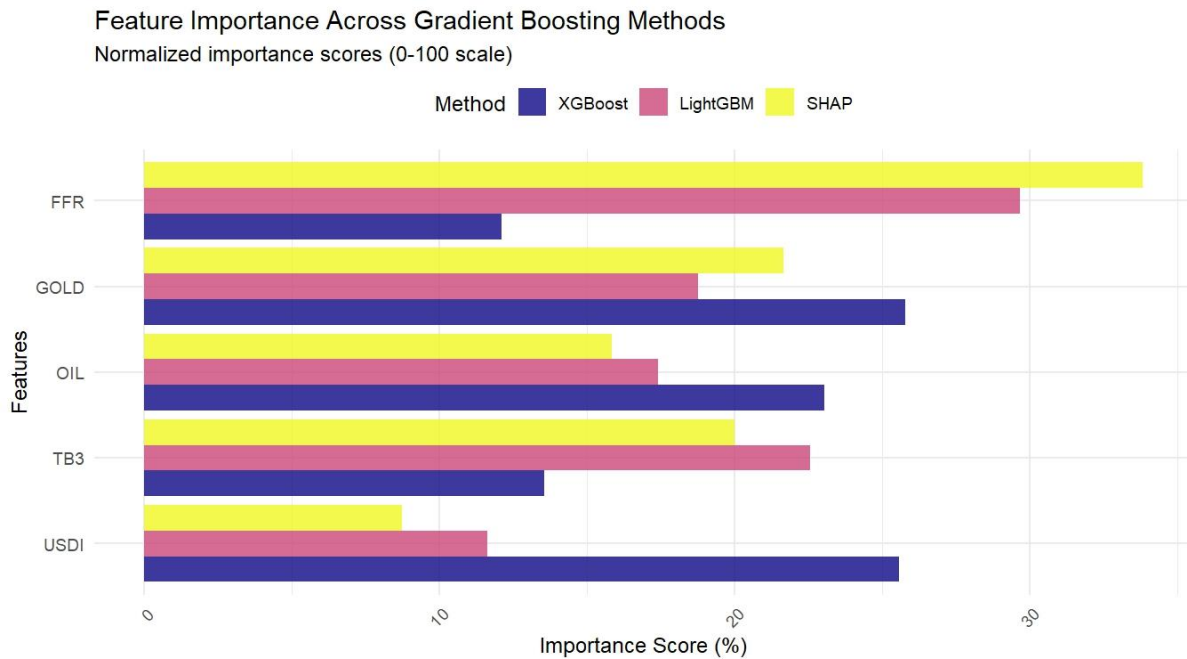


Fig 11: Macroeconomic Feature Importance Across Gradient Boosting Methods

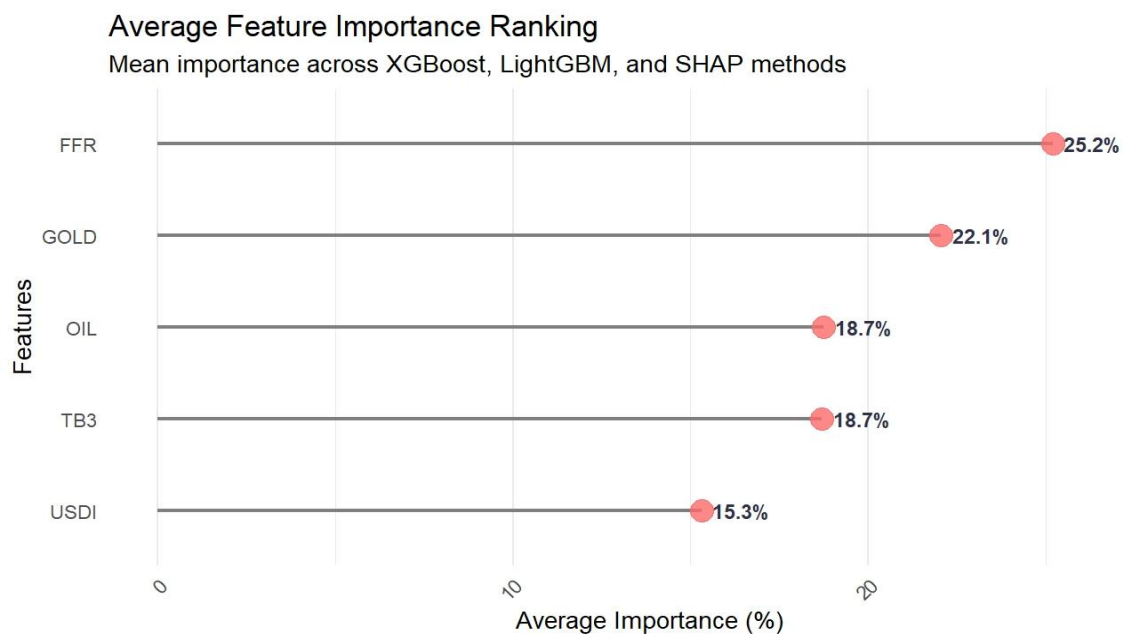


Fig 12: Average Macroeconomic Feature Importance Ranking

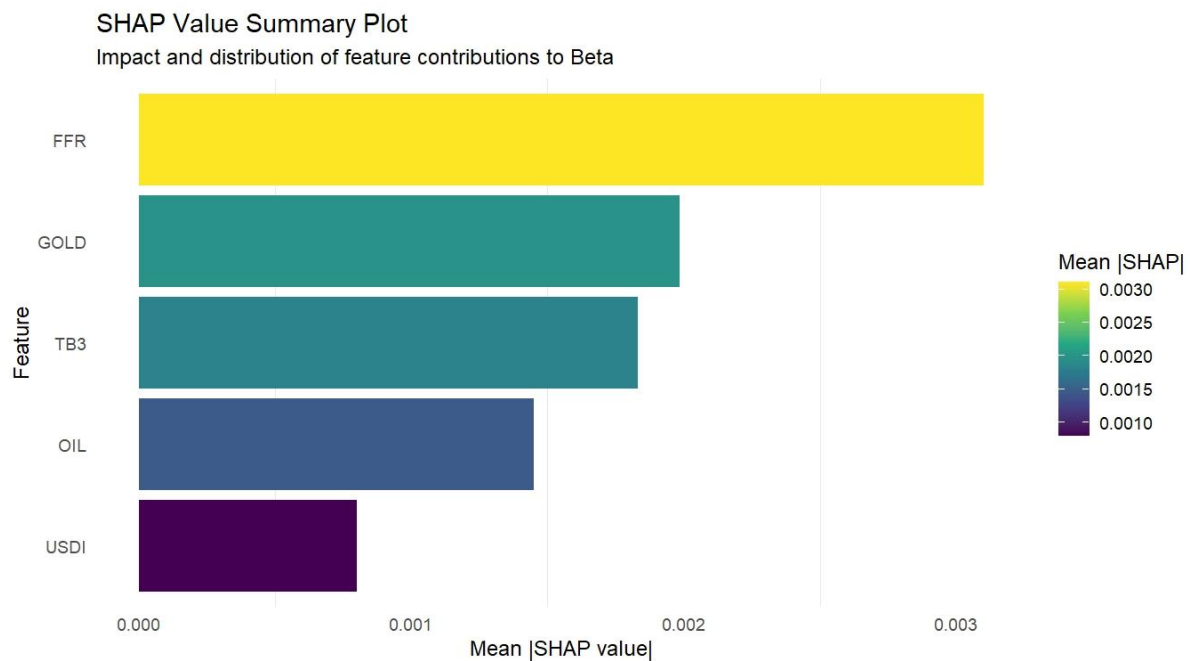


Fig 13: SHAP Value Summary Plot-Macroeconomic Features

The supremacy of the Federal Funds Rate is a profound finding. It positions U.S. monetary policy as a primary driver of cryptocurrency systematic risk. This aligns with financial theory on risk assets: as interest rates rise, the opportunity cost of holding non-yielding speculative assets like Bitcoin increases, potentially leading to outflows and higher volatility (Brière, Oosterlinck, & Szafarz, 2015). Furthermore, tighter monetary policy often signals a contractionary economic phase, driving a "flight to safety" away from risky assets, thereby increasing their perceived systematic risk. The high SHAP value for FFR (Fig. 13) confirms that variations in this rate have a substantial marginal impact on the model's prediction of beta, underscoring its critical role.

The strong showing of Gold (GOLD) and, to a lesser extent, Oil (OIL) is equally significant. Gold is traditionally viewed as a safe-haven asset and a hedge against inflation and macroeconomic uncertainty. cryptocurrency's sensitivity to gold prices suggests that a segment of the market treats it as "digital gold"—a speculative store of value whose risk profile is influenced by the same macroeconomic forces that drive demand for traditional havens (Baur,

Hong, & Lee, 2018). The importance of oil prices likely reflects broader commodity market sentiment and inflation expectations, which indirectly affect investor appetite for alternative assets like cryptocurrencies.

The moderate importance of short-term rates (TB3) reinforces the monetary policy channel, while the relatively lower ranking of the U.S. Dollar Index (USDI) is intriguing. A stronger dollar typically pressures dollar-denominated commodities and risk assets. Its lesser influence here, compared to direct policy rates, may indicate that the mechanism of monetary policy (interest rates) is more critical to crypto investors than its outcome (currency strength) in the short term.

In general, the empirical evidence across blockchain network factors, global financial indices, and macroeconomic and commodity drivers provides a wide-ranging pattern of the determinants of Bitcoin's time-varying beta between 2010 and 2025. By combining results from XGBoost, LightGBM, and SHAP analyses, as well as the evidence from three diverse categories of pictures, and presented in Figures 5 through 13, we observed a dual narrative emerges: Bitcoin's systematic risk is primarily anchored in its intrinsic network fundamentals, while global financial conditions and macroeconomic forces exert secondary yet significant influences.

Figures 5 and 6 clearly illustrate that Bitcoin's systematic risk is most strongly determined by internal blockchain activity, particularly transaction fees (55.2%), number of transactions (19.8%), and hash rate (19%). The SHAP summary plot (Figure 7) further emphasizes that marginal changes in transaction fees have the largest predictive effect on beta. These results suggest that Bitcoin's risk exposure is endogenous to the network itself: congestion and fee escalation signal speculative surges, heightened user demand, and liquidity pressures, all of which amplify systematic co-movement with global risk factors (Chiu & Koepl, 2019). This finding echoes liquidity-adjusted asset pricing theories, which posit that higher transaction costs increase risk premia (Acharya & Pedersen, 2005). For investors, monitoring fee structures and transaction volumes provides forward-looking insight into when Bitcoin may transition from an idiosyncratic diversifier to a pro-cyclical risky asset.

The second layer of influence comes from global financial indices. Figures 8–10 show that the OFR Financial Stress Index (FSI) dominates (42.2%), followed by the VIX (34.3%) and global economic policy uncertainty (23.5%). These results confirm that Bitcoin's beta increases

during periods of market-wide stress and volatility. Contrary to its reputation as a hedge or safe haven, Bitcoin appears to be most exposed during systemic crises, aligning with findings that its diversification benefits collapse under stress (Bouri et al., 2017; Corbet et al., 2020). The SHAP plot (Figure 10) demonstrates that spikes in financial stress and volatility are associated with sharp increases in beta, reinforcing the interpretation that cryptocurrencies are now embedded within broader risk-on and risk-off cycles (Kajtazi & Moro, 2019). For portfolio construction, this implies that risk management strategies must account for Bitcoin’s procyclicality, particularly when designing hedging strategies against equity downturns.

A third set of influences arises from macroeconomic and commodity variables. Figures 11–13 highlight a more distributed ranking of importance, led by the Federal Funds Rate (25.2%), followed by gold (22.1%), oil (18.7%), and short-term interest rates (18.7%), with the U.S. dollar index trailing at 15.3%. Unlike the dominance of network factors, the explanatory power of macro variables is weaker ($R^2 \approx 0.15\text{--}0.17$), but the narrative is economically significant. U.S. monetary policy, captured by the Federal Funds Rate, emerges as the most critical external driver. Higher rates raise the opportunity cost of holding speculative, non-yielding assets like Bitcoin, thereby elevating its beta (Brière, Oosterlinck, & Szafarz, 2015). Gold’s influence reflects Bitcoin’s positioning as “digital gold” in the eyes of investors, with both assets responding to inflationary pressures and safe-haven demand (Baur, Hong, & Lee, 2018). Oil and commodity dynamics add an inflation channel, while the relatively modest role of the dollar index suggests that direct policy instruments matter more than currency outcomes.

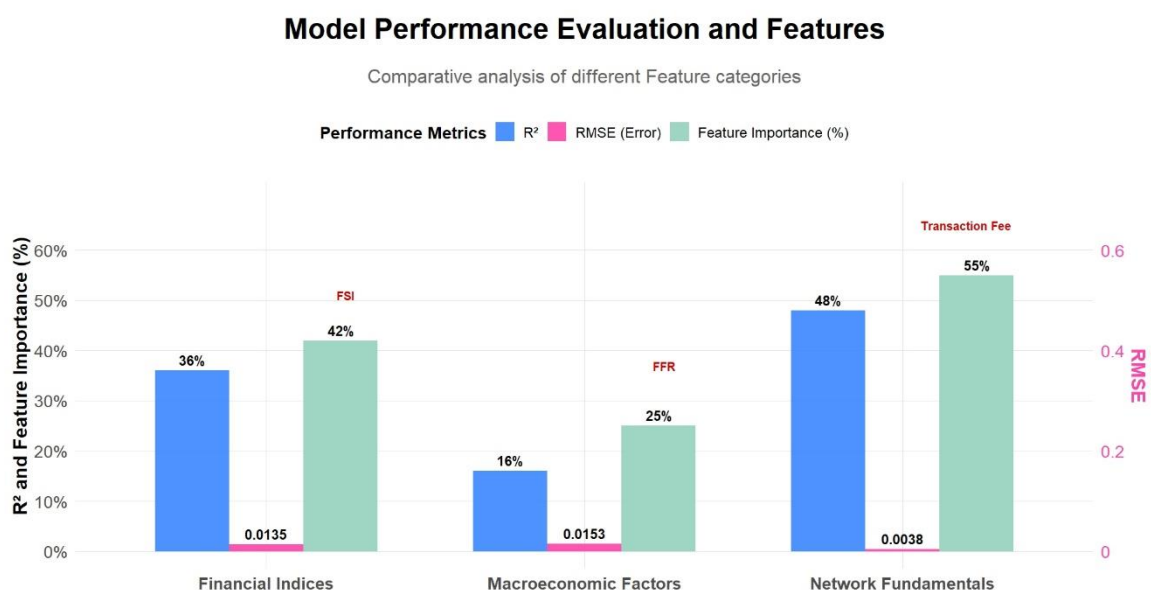


Fig.14: Model Performance evaluation and the importance of features

Figure 14 shows a comparative evaluation of cryptocurrencies beta predictive models across feature categories. When the three categories of features are compared, a clear hierarchy emerges. Network fundamentals explain the bulk of beta variation with high predictive accuracy ($R^2 \approx 0.95$), global financial indices offer moderate explanatory power ($R^2 \approx 0.34$ – 0.39), and macro-commodity factors provide modest contributions. Network fundamentals exhibit the strongest predictive performance ($R^2 = 0.48$), where transaction fees represent the most significant feature (55% importance). In Table 4, error metrics (RMSE) further confirm the superior performance of network-based models (RMSE = 0.0038) compared to financial (RMSE = 0.0135) and macroeconomic (RMSE = 0.0153) approaches. Financial indices demonstrate moderate explanatory power ($R^2 = 0.36$) with the Financial Stress Index (FSI) emerging as the most influential feature (42% importance). Macroeconomic factors show lower predictive capacity ($R^2 = 0.16$) with the Federal Funds Rate (FFR) as the dominant feature (25% importance).

Table 4: Model Performance evaluation metrics

Model Performance by Feature Categories		
Network fundamentals	Financial indices	Macroeconomic factors
RMSE: 0.0038	RMSE: 0.0135	RMSE: 0.0153
R^2 : 0.9486	R^2 : 0.3407	R^2 : 0.1543

This ordering indicates that cryptocurrency’s systematic risk is neither purely speculative nor fully exogenous; instead, it is best understood through a dual-factor framework. First, intrinsic blockchain health—via fees, transactions, and security—anchors risk exposure. Second, external shocks from global stress, volatility, and monetary policy modulate these exposures. This analysis is consistent with financialization theories, which argue that emerging assets and fundamentally new asset class become increasingly tied to systemic cycles over time (Baur et al., 2018).

Moreover, the analysis presented in this section helps resolve previously divergent strands of the literature. Early studies show cryptocurrency's potential as a hedge or safe haven (Dyhrberg, 2016), while others emphasized its speculative nature (Cheah & Fry, 2015) or fundamental adoption dynamics (Peterson, 2018). Several recent evidences show and support conditional safe-haven properties alongside growing financial integration (Bouri et al., 2017; Corbet et al., 2020; Stensås et al., 2019). The findings of this study suggest a blend: while cryptocurrency beta remains predominantly influenced by internal network fundamentals, it is increasingly shaped by external financial and macroeconomic conditions. Accordingly, cryptocurrencies are in a hybrid position in the financial markets, functioning both as speculative technology-based assets and as financialized instruments sensitive to global cycles.

VIII. Discussion

The results of this study reveal the limits of traditional constant-beta frameworks in explaining cryptocurrency returns and provide evidence for the benefits of a dynamic, multifactor approach. The findings in summary statistics suggest that cryptocurrencies exhibit extreme heterogeneity in their distributional properties, with wide ranges of returns and frequent fat-tailed shocks. Such characteristics are difficult to reconcile with static models that assume stable risk-return trade-offs. The dispersion of data across assets, coupled with an unbalanced panel, further motivates the adoption of advanced econometric tools capable of capturing evolving exposures. Correlation patterns strengthen this point by showing clusters of tightly interconnected assets like Bitcoin, Ethereum, and Litecoin alongside more idiosyncratic entrants such as Solana and Tron, suggesting that systemic and asset-specific risks coexist and evolve over time.

The inadequacy of constant betas becomes clear when examining static CAPM regressions. Persistent negative alphas, low explanatory power, and wide scatter around fitted values indicate that systematic market risk explains little of the cross-sectional variation in returns. Coins with high market sensitivity, such as Solana and XRP, do not earn proportionally higher expected returns, pointing to inefficiencies inconsistent with classical finance theory. For many digital assets, role of dominating idiosyncratic volatility show the limited applicability of single-factor risk pricing models. These findings align with broader evidence that static risk-return frameworks are poorly suited to cryptocurrencies, which are instead shaped by technological narratives, adoption cycles, and fluctuating liquidity conditions.

Allowing betas to vary with the time reveals a more realistic nature of systematic risk dynamics. Bitcoin's gradual upward trend in beta reflects its ongoing integration into traditional financial markets, whereas Ethereum's sensitivity surged during the DeFi and NFT expansion before declining alongside speculative activity. Solana exhibited episodic spikes in market sensitivity, corresponding to phases of market exuberance and subsequent retrenchment, while Monero followed a hump-shaped trajectory influenced by its early days' regulatory environment. These shifting pathways illustrate how technological features, investor sentiment, and liquidity cycles interact with systematic risk exposure, supporting the relevance of conditional models in which betas evolve alongside economic and behavioral regimes.

Predictive analysis using machine learning approaches further explains the drivers of these variations. For Bitcoin, network fundamentals, such as transaction fees, transaction volume, and hashrate account for a substantial share of beta fluctuations, suggesting that blockchain activity serves as a leading indicator of systematic risk. Nevertheless, global financial stress measures and monetary policy also exert considerable influence, with Bitcoin displaying increased pro-cyclicality during crises and extreme sensitivity to changes in the cost of capital. This dual dependency indicates that cryptocurrencies are not isolated technologies but assets oscillating between innovation-led and finance-led risk regimes. The application of XGBoost and LightGBM, complemented by SHAP values, confirms the nonlinear interplay between endogenous and exogenous forces in shaping crypto risk.

From a portfolio perspective, these findings evidently highlight the necessity of moving beyond static asset allocations. Dependence on CAPM-style constant betas results in an underestimation of systemic exposure during critical episodes, while neglecting correlation clustering leaves portfolios vulnerable to sudden contagion. Effective diversification requires monitoring both dynamic betas and evolving interdependencies, with particular attention to network fundamentals as forward-looking risk signals. For investors, high-beta assets such as Solana and XRP may serve to amplify market exposure, while relatively stable assets like Bitcoin and Monero offer conditional hedging potential rather than absolute safe-haven properties. These results emphasize that cryptocurrency portfolios require dynamic management, adapting to shifting risk loadings and acknowledging the hybrid role of these assets as both speculative technologies and financial instruments.

IX. Conclusion and Portfolio Implications

In this study, we examine the understanding of systematic risk in cryptocurrency markets by moving beyond the static assumptions in traditional asset pricing models. Traditional approaches and theories, such as the CAPM and APT provide foundational insights into the risk–return trade-off. However, they remain limited when applied to digital assets that are characterized by extreme volatility, heterogeneous adoption patterns, and evolving technological infrastructures. Motivated by the inadequacy of constant-beta formulations, the research aimed to develop and test a dynamic, cryptocurrency-specific multifactor APT framework capable of capturing time-varying betas that reflect blockchain fundamentals, investor sentiment, and macro-financial conditions. In doing so, this project addresses three research questions: how systematic risk in cryptocurrencies evolves over time, which factors drive this variation, and what implications such dynamics hold for portfolio construction and risk management?

Empirical analysis confirms that cryptocurrencies deviate substantially from the assumptions of traditional models. Static CAPM results reveal persistent inefficiencies, as market betas fail to explain the cross-sectional distribution of returns and high-beta assets do not consistently deliver commensurate compensation. Allowing for time-varying betas, however, uncovers meaningful dynamics: Bitcoin’s gradual rise in market sensitivity, Ethereum’s cyclical swings during periods of technological enthusiasm, Solana’s sharp boom-and-bust exposures, and Monero’s regulatory-driven fluctuations all illustrate the state-dependent and episodic nature of risk in digital assets. The integration of machine learning methods further highlights that blockchain network activity—particularly transaction counts, fees, and hashrate—constitutes the dominant driver of systematic risk, while global financial stress and monetary policy variables modulate exposures during crises. These findings provide strong evidence that cryptocurrency risk is endogenous to blockchain health but amplified or dampened by exogenous financial shocks.

We present the practical implications of the findings of this study for investors and portfolio managers as follows: Static beta tilts or one-size-fits-all diversification strategies are ineffective in markets where systematic risk is dynamic and heterogeneous across assets. Instead, effective portfolio design requires monitoring shifting correlations and identifying network-based signals that foreshadow regime changes in betas. Bitcoin and Monero may serve as relative stabilizers during certain phases, while high-beta assets such as Solana or XRP offer opportunities for risk-seeking allocations, provided investors account for their cyclical

sensitivity to global shocks. In this way, the study demonstrates that cryptocurrencies must be managed as a hybrid class of speculative technologies and financial instruments, rather than as a homogenous risk category.

The contribution of this research lies in developing a dynamic linear APT framework tailored to the cryptocurrency ecosystem, complemented by state-space econometric methods and interpretable machine learning. By constructing a long-span dataset of blockchain, market, and macro-financial variables covering 15 major assets, the study provides one of the most comprehensive empirical analyses of systematic risk in digital markets to date. It bridges microeconomic theory, behavioral dynamics, and computational techniques to reveal the endogenous and exogenous forces shaping crypto risk exposures.

However, the reliance on daily data excludes intraday microstructure effects, while the use of historical blockchain and sentiment variables may understate the role of emerging decentralized applications and cross-chain innovations. Furthermore, the machine learning models, while powerful in ranking factor importance, do not provide full structural interpretations of causal channels. These limitations highlight several avenues for future work, including high-frequency modeling, network-based contagion analysis, and integration of decentralized finance (DeFi) and stablecoin dynamics into broader risk frameworks.

In general, this study demonstrates that systematic risk in cryptocurrency markets is highly dynamic, jointly shaped by blockchain fundamentals, global macro-financial cycles, and investor sentiment. Time-varying betas provide a more accurate and policy-relevant lens through which to analyze these assets, offering investors critical insights into risk management and portfolio design. By situating cryptocurrencies at the intersection of technological innovation and financial integration, the findings underscore both the opportunities and vulnerabilities that define this evolving market.

References

- Acharya, V. V., & Pedersen, L. H. (2005). Asset pricing with liquidity risk. *Journal of Financial Economics*, 77(2), 375–410.
- Adrian, T., & Brunnermeier, M. K. (2016). CoVaR. *American Economic Review*, 106(7), 1705–1741.
- Ang, A., & Chen, J. (2007). CAPM over the long run: 1926–2001. *Journal of Empirical Finance*, 14(1), 1–40.

- Athey, S., Parashkevov, I., Sarukkai, V., & Xia, J. (2016). Bitcoin pricing, adoption, and usage: Theory and evidence. *Unpublished manuscript*.
- Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4), 1593–1636.
- Baur, D. G., Hong, K., & Lee, A. D. (2018). Bitcoin: Medium of exchange or speculative assets? *Journal of International Financial Markets, Institutions and Money*, 54, 177–189.
- Biais, B., Bisiere, C., Bouvard, M., & Casamatta, C. (2019). The blockchain folk theorem. *Review of Financial Studies*, 32(5), 1662–1715.
- Borri, N. (2019). Conditional tail-risk in cryptocurrency markets. *Journal of Empirical Finance*, 50, 1–19.
- Borri, N., & Shakhnov, K. (2023). Cryptomarket discounts. *Journal of International Money and Finance*, 139, 102963.
- Bouri, E., Molnár, P., Azzi, G., Roubaud, D., & Hagfors, L. I. (2017). On the hedge and safe haven properties of Bitcoin: Is it really more than a diversifier? *Finance Research Letters*, 20, 192–198.
- Bouri, E., Shahzad, S. J. H., Roubaud, D., & Kristoufek, L. (2020). Cryptocurrencies as hedges and safe havens for US equity sectors. *Quarterly Review of Economics and Finance*, 75, 294–307.
- Brière, M., Oosterlinck, K., & Szafarz, A. (2015). Virtual currency, tangible return: Portfolio diversification with bitcoin. *Journal of Asset Management*, 16(6), 365–373.
- Brooks, R. D., Faff, R. W., & Lee, J. H. (1992). The form of time variation of systematic risk: Some Australian evidence. *Applied Financial Economics*, 2(4), 191–198.
- Cai, C. W. (2019). Disruption of financial intermediation by blockchain: A review on cryptocurrency and ICOs. *Review of Financial Economics*, 37(2), 146–153.
- Cheah, E. T., & Fry, J. (2015). Speculative bubbles in Bitcoin markets? An empirical investigation into the fundamental value of Bitcoin. *Economics Letters*, 130, 32–36.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794).
- Chiu, J., & Koeppl, T. V. (2019). Blockchain-based settlement for asset trading. *Review of Financial Studies*, 32(5), 1716–1753.
- Cong, L. W., He, Z., & Li, J. (2021). Decentralized mining in centralized pools. *Review of Financial Studies*, 34(3), 1191–1235.
- Cong, L. W., Li, Y., & Wang, N. (2021). Tokenomics: Dynamic adoption and valuation. *Review of Financial Studies*, 34(3), 1105–1155.

- Corbet, S., Lucey, B., Urquhart, A., & Yarovaya, L. (2019). Cryptocurrencies as a financial asset: A systematic analysis. *International Review of Financial Analysis*, 62, 182–199.
- Corbet, S., Lucey, B., Urquhart, A., & Yarovaya, L. (2020). Cryptocurrencies as a financial asset: A systematic analysis. *International Review of Financial Analysis*, 71, 101506.
- Corbet, S., Meegan, A., Larkin, C., Lucey, B., & Yarovaya, L. (2018). Exploring the dynamic relationships between cryptocurrencies and other financial assets. *Economics Letters*, 165, 28–34.
- Demir, E., Gozgor, G., Lau, C. K. M., & Vigne, S. A. (2018). Does economic policy uncertainty predict the Bitcoin returns? *Finance Research Letters*, 26, 145–149.
- Dyhrberg, A. H. (2016). Bitcoin, gold and the dollar—A GARCH volatility analysis. *Finance Research Letters*, 16, 85–92.
- Engle, R. F. (2016). Dynamic conditional correlation: A simple class of multivariate GARCH models. *Journal of Business & Economic Statistics*, 20(3), 339–350.
- Fabozzi, F. J., & Francis, J. C. (1978). Beta as a random coefficient. *Journal of Financial and Quantitative Analysis*, 13(1), 101–116.
- Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- Fama, E. F., & MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3), 607–636.
- Feng, W., Wang, Y., & Zhang, Z. (2018). Can cryptocurrencies be a safe haven? A tail risk perspective analysis. *Applied Economics*, 50(44), 4745–4762.
- Ferson, W. E., & Harvey, C. R. (1991). The variation of economic risk premiums. *Journal of Political Economy*, 99(2), 385–415.
- Ferson, W. E., & Harvey, C. R. (1999). Conditioning variables and the cross section of stock returns. *Journal of Finance*, 54(4), 1325–1360.
- Fraser, P., Hamelink, F., Hoesli, M., & Macgregor, B. (2004). Time-varying betas and the cross-sectional return–risk relation: Evidence from the UK. *European Journal of Finance*, 10(4), 255–276.
- Han, Y., Liu, X., & Tse, Y. (2022). Momentum in cryptocurrency returns. *Journal of Futures Markets*, 42(2), 235–252.
- Harvey, C. R. (1989). Time-varying conditional covariances in tests of asset pricing models. *Journal of Financial Economics*, 24(2), 289–317.
- Hrdlicka, C. (2022). Trading volume and time varying betas. *Review of Finance*, 26(1), 79–116.

- Jagannathan, R., & Wang, Z. (1996). The conditional CAPM and the cross-section of expected returns. *Journal of Finance*, 51(1), 3–53.
- Jensen, M. C. (1968). The performance of mutual funds in the period 1945–1964. *Journal of Finance*, 23(2), 389–416.
- Jo, H., Park, H., & Shefrin, H. (2020). Bitcoin and sentiment. *Journal of Futures Markets*, 40(12), 1861–1879.
- Kajtazi, A., & Moro, A. (2019). The role of Bitcoin in well diversified portfolios: A comparative global study. *International Review of Financial Analysis*, 61, 143–157.
- Katsiampa, P. (2019). Volatility co-movement between Bitcoin and Ether. *Finance Research Letters*, 30, 221–227.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... Liu, T. Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30, 3146–3154.
- Li, T., & Wang, Y. (2017). The technology and economic determinants of cryptocurrency exchange rates: The case of Bitcoin. *Decision Support Systems*, 95, 49–60.
- Lintner, J. (1965). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics*, 47(1), 13–37.
- Lintner, J. (1975). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. In *Stochastic optimization models in finance* (pp. 131–155). Academic Press.
- Liu, Y., & Tsyvinski, A. (2021). Risks and returns of cryptocurrency. *Review of Financial Studies*, 34(6), 2689–2727.
- Liu, Y., Tsyvinski, A., & Wu, X. (2019). Common risk factors in cryptocurrency. *Journal of Finance*, 74(6), 2489–2530.
- Liu, Y., Tsyvinski, A., & Wu, X. (2022). Common risk factors in cryptocurrency. *Journal of Finance*, 77(2), 1133–1177.
- Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774.
- Makarov, I., & Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2), 293–319.
- Mankiw, N. G., & Shapiro, M. D. (1984). Risk and return: Consumption versus market beta (No. w1399). *National Bureau of Economic Research*.
- Pagnotta, E., & Buraschi, A. (2018). An equilibrium valuation of bitcoin and decentralized network assets. *Available at SSRN 3142022*.

- Peterson, T. (2018). Metcalfe's law as a model for Bitcoin's value. *Alternative Investment Analyst Review*, 7(2), 9–18.
- Petris, G., Petrone, S., & Campagnoli, P. (2009). *Dynamic linear models with R*. Springer.
- Ramos, S., Pianese, F., Leach, T., & Oliveras, E. (2021). A great disturbance in the crypto: Understanding cryptocurrency returns under attacks. *Blockchain: Research and Applications*, 2(3), 100021.
- Raponi, V., Robotti, C., & Zaffaroni, P. (2020). Testing beta-pricing models using large cross-sections. *Review of Financial Studies*, 33(6), 2796–2842.
- Ross, S. (1976). The arbitrage pricing theory. *Journal of Economic Theory*, 13(3), 341–360.
- Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3), 425–442.
- Stensås, A., Nygaard, M. F., Kyaw, K. S., & Treepongkaruna, S. (2019). Can Bitcoin be a diversifier, hedge or safe haven tool? *International Review of Financial Analysis*, 63, 322–330.
- Treynor, J. L. (1962). Toward a theory of market value of risky assets. *Unpublished manuscript*. (Final version published in 1999, in *Asset Pricing and Portfolio Performance: Models, Strategy and Performance Metrics*, R. A. Korajczyk, Ed., pp. 15–22).
- Umlandt, D. (2023). Score-driven asset pricing: Predicting time-varying risk premia based on cross-sectional model performance. *Journal of Econometrics*, 237(2), 105470.
- Wang, Q., & Chong, T. T. L. (2021). Factor pricing of cryptocurrencies. *North American Journal of Economics and Finance*, 57, 101348.
- Woebbecking, F. (2021). Cryptocurrency volatility markets. *Digital Finance*, 3(3), 273–298.
- Yermack, D. (2015). Is Bitcoin a real currency? An economic appraisal. In D. Lee (Ed.), *Handbook of Digital Currency* (pp. 31–43). Academic Press.